All the Headlines that Are Fit to Change*

Tin Cheuk Leung[†]
Wake Forest University

Koleman Strumpf[‡]
Wake Forest University

March 2023

Abstract

We study how the New York Times changes headlines after an article has been published. We track the 35,000 articles on the newspaper's home page between February 2021 and April 2022, and find that 10% have an immediate headline change while another 11% undergo an A/B test where two titles are trialed against one another over a short time window. We also collect every tweet of an article in our data. We first look at which articles get a headline change of some form. Article type matters: popularity on Twitter, hard news, negative sentiment headline, and mentions from more partisan Twitter accounts are associated with headline changes. Another factor is social pressure, negative tweets from users connected to the reporter or the newspaper, which leads to more immediate headline changes and to a lesser extent to A/B tests. Next we examine how headline changes impact an article's performance metrics. After accounting for selection, we find that headline changes increase an article's popularity. There is also evidence that headline changes increase the negative sentiment and liberal political slant when it is tweeted. We also analyze headline changes at the Wall Street Journal for a shorter period, and find comparable behavior. Our results have implications for distinguishing supply and demand driven models of media bias, how digitization may foster media consolidation, and conditions under which economic or partisan motives drive newspaper decision-making.

JEL Classification: D83, L82
Keywords: A/B Testing, Media Economics, Machine Learning, Social Media

*We would like to thank Luis Aguiar, John Dalton, Simone Galperti, Lisa George, Aeimit Lakdawala, Christian Peukert, David Rothschild, and seminar participants at CUHK-Zhejiang University Workshop on Media Economics, Penn Carey (Economics of Digital Services Research Symposium), CREED, Microsoft Research Lab - New York City, NYC Media Seminar (Columbia and CUNY Hunter), SEA, Virtual Digital Economy (VIDE) seminar, William and Mary for their comments, Sean Anderson of the WFU HPC Facility, Tom Lam, Julian TszKin Chan for their help on coding, Tom Cleveland on sharing his NYT headline data and code for the NYT scraper, the Ground News on sharing its Blindspotter data. We also gratefully acknowledge financial help from Center for Technology, Innovation and Competition (CTIC) from the University of Pennsylvania and the Knight Foundation. The usual disclaimer applies.

[†]Contact: Department of Economics, Kirby Hall, Wake Forest University, Box 7505, Winston-Salem, NC 27109. Email: leungtc@wfu.edu

[‡]Email: strumpks@wfu.edu

"And now it's alright, it's okay
And you may look the other way
We can try to understand
The New York Times' effect on man"
from The Bee-Gees song, Stayin' Alive

1 Introduction

In March of 2020 during the first days of the coronavirus outbreak in the US, the New York Times (NYT) altered the online headline three times for an article on the president's relief plan. "Democrats Block Action on $1.8 Trillion Stimulus" was changed to "Democrats Block Action on Stimulus Plan, Seeking Worker Protections" and finally to "Partisan Divide Threatens Deal on Rescue Plan," while the text of the article remained largely the same (later there was yet another headline change). President Trump tweeted in response: "The New York Times changed headlines 3 times in order to satisfy the Radical Left," "very unfair," "Fake & Corrupt News," "very dangerous for our Country!" "Fake News," "ALL THE NEWS THAT'S NOT FIT TO PRINT."

This anecdote raises several questions. How often and when are headlines changed? How does the newspaper decide which headline is best? Why are they changed? Does the newspaper change headlines for economic or ideological reasons? What is the effect of these changes? In this paper we will investigate these questions. In addressing them in full or in part, we will highlight an underappreciated aspect of the transition from physical to digital news: the ability to modify stories after they are published.[1]

Headline changes are important for many reasons.[2] Headlines play a central role in which of the many articles on the newspaper's home page are read. The psychology

[1]Newspapers do change print headlines between editions. For example, on 19 August 2019 the New York Times changed a headline on mass shootings from "Trump Urges Unity vs. Racism" to "Assailing Hate But Not Guns" between print editions. Still headline changes in physical print are far harder to do, and allow only a single change with little ability to specify the time when the change is made. In addition, the print edition often uses a different headline than the online version.

[2]While journalists consider a headline to be separate from the underlying article, few readers see such a distinction. The headlines are often what attracts (or does not attract) a reader to engage with the article. It is worth noting some newspaper's identity is entwined with their headlines, with two notable ones being the New York Post's comic front-page (New York Post, 2007) and Variety's slanguage (riffs on shortened industry slang). Also some of the most memorable stories in journalism are due to headline gaffes: "Dewey Defeats Truman," "All Titanic Passengers are Safe," "Kerry's Choice: Dem Picks Gephardt as VP Candidate."

literature also shows headlines impact how people recall articles and that misleading headlines cause them to mis-remember article content (Ecker et al., 2014). In short, first impressions matter (Konnikova, 2014). Another reason headline changes matter is they are increasingly being used throughout the media industry. Chartbeat, a leading provider of analytical tools for publishers, says 80% of the top US media publications use a form of headline testing. It is also used in 68 countries on six continents (Ray, 2022). Understanding the causes and effects of this increasingly common tool will give insights into the media industry. Headline changes can play a role in the industrial organization of the news media. Headline changes are best done at scale, with a larger readership providing more insights into how to change headlines. They also require technical expertise. To the extent headline testing is a tool that benefits newspapers, whether it is profitability or some other motive, this is a novel way that digitization pushes to consolidation and perhaps disadvantages smaller, local newspapers. Finally, unlike other content change such as corrections, newspapers do not seem to give a full history of the headline changes (this is true for the NYT app and NYT webpage). Even an attentive reader would have no idea whether a headline was changed, and if so which version they receive. This can lead to reader confusion and calls into question the use of newspapers as a fixed-in-amber source.

We analyze each of the 35,164 articles which appeared on the NYT home page at some point over the period February 2021 - April 2022 with the exception of July 2021.[3] Along with information from the home page we add variables from the NYT API, and in total we download the headline, a snippet of the article text, url, initial posting time, internal ranking in terms of views and shares, as well as metadata such as article tone and section it appeared in. There is also a unique identifier which allows us to track the entire history of headline changes (our scraper runs every five minutes between February - June 2021, and every one minute between August 2021 - April 2022). We also score each headline using standard sentiment analysis algorithms. To get a better sense of the impact of headline changes on the articles, we collect every tweet which shares the url of an article in our database (we provide evidence that twitter users who tweet NYT stories are comparable to the broader NYT readership). For each tweet we know the popularity (number of tweets and retweets) and can again use sentiment analysis on the text. For each Twitter author we have a measure of their partisanship (based on the composition of

[3]We missed one month of data due to the transition between servers hosting the scraper.

news sources they tweet) and for many we know their location (country, state and city). And finally we can calculate each tweet's political slant, how liberal it is, using tweets from politicians and machine learning based on usage of key words and the politicians' political affiliation (Democrat or Republican).

Our first results involve how the NYT make online headline changes, which the paper has argued might reflect new information or may help with search engine optimization (Symonds, 2017). They employ two approaches: they either change to a new headline or they employ A/B tests in which two headlines are shown to observationally equivalent readers and based on their response one of them is selected at the end of the experiment period (readers appear to be identified using cookies and user agents, so to avoid being stuck in a single group our scraper uses a randomly-generated user-agent and never sends cookie headers).[4] About 11% of articles have an A/B test. Another approach is to change to a new headline without a test (an immediate headline change), which happens with another 10% of articles.

Our empirical analysis is in two parts. First we look at what factors lead an article to get a headline change. Article type matters: more popular articles (those more quickly placed on the NYT homepage or which get more tweets when the article is initially published), hard news, negative sentiment headline, and mentions from more partisan Twitter accounts are associated with both types of headline change. Another factor is social pressure, voice of reader dissatisfaction with an article (see Gans, Goldfarb, and Lederman, 2021). Our measure for social pressure is negative tweets from users connected to the reporter or the newspaper (a user is connected if the article's reporter or newspaper accounts follow the user on Twitter). We find such negative tweets lead to more immediate headline changes and to a lesser extent A/B tests. There is little effect from tweets mentioning either the reporter or newspaper if they come from non-connected users or from users located just in the US. Together these results suggest external factors play an important role in the newspaper's decision about which headlines to change.

Next we examine how headline changes impact an article's performance metrics. Since it is not random which articles get headline changes, non-changed articles are an inappropriate counterfactual. Instead we use two approaches to address the selection

[4]The NYT has acknowledged, and external sources have verified, using A/B tests since at least 2015 (Arak and Kaji, 2017; Reisman, 2016b). The newspaper's senior editors in charge of headlines say that such tests are used when "a story isn't performing as well as expected" with the goal of finding a title which attracts the most readers (Bulik, 2016).

issue. First, we employ the propensity score matching (PSM) method to pair articles with headline changes to articles without headline changes based on key observed covariates. Second, we use the number of changes of other articles in the same newspaper section as an instrument. The intuition for the choice of instrument is that headline changes occur in clumps, likely due to scale economies. After accounting for selection, we find that both type of headline change increases an article's popularity (number of tweets). We also find evidence that the tweets sharing the article also become more negative in their sentiment and more pro-Democrat in their political slant in some specifications, especially among articles which undergone A/B tests. In total the results are mainly consistent with using headline changes to increase article popularity by spreading negative sentiment and political slant.

Finally we perform the same analysis on the Wall Street Journal, the right of center analogue to the NYT. While currently the analysis is restricted to a four month period (October 2022 to January 2023), we find the Wall Street Journal uses the same two types of headline changes and enacts them at the same rate as the NYT. We also find similar impacts in terms of an elevation in the number of tweets an article receives and the negative sentiment of those tweets following a headline change in some set of our results. This suggests that the patterns we find above might extend beyond the NYT to a broader set of newspapers.

There are some broader lessons from our analysis. First, headline changes have implications for market structure. Learning which headlines perform better is best done at scale, with a larger newspaper able to garner more views and more quickly determining a headline's efficacy. A smaller newspaper may not be able to do this until the article is stale. This would be a novel channel by which digitization, a necessary ingredient to headline changes, leads to market consolidation. Second, it provides a useful environment to test leading theories of media markets. For example, while newspapers have readily identifiable biases it is unclear whether supply side (newspaper preferences) or demand side (consumer preferences) drive this. This is a challenging empirical ask, which are able to address since we observe consumer reactions when newspapers change headlines and their resulting political slant. In our analysis of the Wall Street Journal, we find consumers are behaving at odds with the predictions of demand side model. Finally, the headline changes can help identify the newspaper objective function. As the initial anecdote illustrates, papers can have partisan aims. But they are also interested in profits.

While both motives manifest in some of the estimates, the majority of the estimates are consistent with profit-seeking. The main effect of headline changes is to increase reader engagement as proxied through Twitter activity. An important extension would be to see if such activity leads to increases in subscriptions, which the NYT has stated is its main financial goal for the foreseeable future.

An important contribution is our study of A/B tests in the wild. Prior studies of A/B tests have focused on the perspective of the experimenter. While this has certain advantages such as an ability to run additional experiments, it raises questions about external validity due to response bias such as social desirability bias or the Hawthorne effect. These concerns have grown especially acute as firms have been harshly criticized for running A/B tests. For example, LinkedIn's 2015-2019 A/B test experiment on the importance of weak social ties in yielding job offer was subject to a media storm when the underlying research was published in 2022 (Rajkumar et al., 2022). We have the perspective of the experimental subject (we passively collect data as if we were an ordinary viewer) which may more accurately reflect typical A/B tests.

In the revised draft we will make several additions that will allow us to more precisely answer these questions. First, we will have additional data. We have continued scraping the NYT API (and WSJ) and we will have data from February 2023 to present. Second, we have access to data from Crowd Tangle, a subsidiary of Meta. This will allow us to do the same analysis we have conducted on Twitter with Facebook. Third, we will investigate the connection of headline changes to Twitter bots. Twitter bots are automated programs designed primarily to amplify (re-tweet or like) certain kinds of messages, primarily to promote a particular ideological viewpoint (some governments have been found to utilize extensive bot networks). If the NYT headlines increase or dampen such bots, this will shed light on the partisan motive. We have access to a site which uses machine learning to classify every twitter account as a bot or not. Finally, we would like to look at other major media sources. Dozens of other media sites, such as CNN, USA Today and ABC, are known to change online headlines directly or based on an A/B test (Reisman, 2016a; Jiang, Martin, and Wilson, 2019). Seeing whether the patterns we observed to the NYT and WSJ carry over to other sources will be an important next step.

The remainder of the paper is as follows. Section 2 reviews the related literature, which is followed by a conceptual framework (Section 3). Then we describe the data in Section 4, and present our empirical results in Sections 5 and 6. The last section concludes.

2 Literature

Our paper contributes to the growling literature of the relationship between social media and news. Several papers study the competition for attention between newspapers and social media (de Corniére and Sarvary, 2022; Jeon and Nasr, 2016), the impact of social media on fake news (Allcott and Gentzkow, 2017), and the impact of social media on news production by mainstream media (Cagé, Hervé, and Mazoyer, 2022; Hatte, Madinier, and Zhuravskaya, 2022). Our paper contributes to the literature in several ways. First, we use tweets as a proxy of an article's popularity. Second, using data from Twitter, we are able to quantify the impacts of headline changes on readers' sentiment and liberal political slant through textual analysis and machine learning methods. Lastly, our results suggest that pressure from social media, in particular from users connected to the reporter or the newspaper, can be one of the factors behind news production decisions such as headline changes.

This paper also contributes to the study of media bias. Groseclose and Milyo (2005) are among the first studies to systematically document media bias in a large scale. They find that all but two (*Fox News' Special Report* and the *Washington Times*) news outlets received scores to the left of the average member of Congress, while NYT received scores far to the left of center. Many studies have since analyzed several drivers of media bias, including demand for media bias (Huang, Meng, and Weng, 2022; Gentzkow and Shapiro, 2010; Martin and Yurukoglu, 2017), supply of media bias (Cagé et al., 2022; Martin and McCrain, 2019), professional norms (Baum and Groeling, 2008; Shapiro, 2016), journalists' ideology (Boxell and Conway, 2022), and the level of market competition competition (Gentzkow and Shapiro, 2008; Qin, Strömberg, and Wu, 2018). Several papers also study the historical evolution of partisan news sources (Gentzkow, Glaeser, and Goldin, 2006; Hirano and Snyder, 2020) and the importance of fake news and partisan news in a typical American's news diet (Allen et al., 2020; Muise et al., 2022). Our results suggest that the use of technology might exacerbate media bias.

There is a growing literature on the impact of the introduction of new media technologies on political particiapation, with evidence from newspapers (Gentzkow, Shapiro, and Sinkinson, 2011; Snyder and Strömberg, 2010), radio (Strömberg, 2004; Wang, 2022), television (Angelucci and Cagé, 2019; Angelucci, Cagé, and Sinkinson, 2020; Gentzkow, 2006), and internet (Boxell, Gentzkow, and Shapiro, 2018; Campante, Durante, and

Sobbrio, 2018; Falck, Gold, and Heblich, 2014). This paper provides new insights on the impact of social media in polarization.

Lastly, this paper contributes to our understanding of the impact of technology in general and in the media industry. Athey and Luca (2019) documents the use of various statistical tools made possible by digital technology, including A/B testing, in Tech companies. Previous studies have examined the impact of technology/statistical tools in the media industry such as how A/B testing algorithms are changing workflow and headline writing practices (Hagar and Diakopoulos, 2019), the returns to data and informational externalities associated with algorithmic recommendations relative to human curation in online news (Claussen, Peukert, and Sen, 2020), and how NYT expansion of its home delivery affects local newspaper circulation (George and Waldfogel, 2006). Our results shed light on the use and the impact of digital technology (headline switch and A/B tests) on articles' headline at the NYT.

3 Conceptual Framework

To organize ideas, we should think about headlines in two steps. In the first, the newspaper decides whether to change it (and how to do so). In the second, we can evaluate the impact of the headline change. We will discuss each of these in turn.

When the NYT decides to change a headline, there are a variety of possible motives. We will focus on two. The first is economic: the headline could attract more reader engagement, which in turn could lead to higher profits through additional subscriptions or higher advertising rates. A second reason is more partisan: the headlines could be used to shape the public discussion of a news story (persuasion) or to attract certain kinds of readers (say ones of a particular ideology), as in President Trump's argument in the introduction. These motives are not necessarily in conflict, e.g. the new headline could attract new readers of a certain ideological tilt. But they might be, with the newspaper perhaps foregoing some revenue to make a political point or conversely to attract a wide range of readers including those of a different ideology. In many ways, this is analogous to the choice of movie studios to focus on economics (blockbusters) or art (prestige films, which typically have a low box office).

A related choice is the process for changing headlines. The newspaper could simply change to a new title (an immediate headline change), or instead it could run an experiment (an A/B test) to compare the incumbent title against an alternative. The latter has the advantage of seeing which title has a better outcome (variously defined), but could cause confusion or even resentment if it becomes commonly known. Recall the backlash against companies engaging in price discrimination such as Amazon with DVDs in 2000, Orbitz steering Mac user to higher priced hotels in 2012, or Princeton Review's charging differential prices based on an online users zipcode.

The other step is to evaluate the impact of the change. The NYT has internal metrics such as the actual number of page views and shares, but only makes public the relative ranking of articles and only the top twenty ranked articles at a daily or even less granular frequency. To overcome this we utilize Twitter data, which can be used to measure user engagement. It also has several other advantages like being able to characterize Twitter users partisanship, looking at their location, and also to see how popular a tweet is in terms of retweets or likes.

4 Data

Our data come from several sources. We use a scraper to collect our main headline data from the NYT, and complement the data with tweets data from the Twitter API and news diet data of those Twitter accounts from Ground News Blindspotter.

4.1 NYT Headlines

We have a scraper which collects data from the NYT, and ran every five minutes from 23 February 2021 through 30 June 2021, and every minue from July 28 2021 through April 2022. On each cycle it collects all articles on the NYT home page, `https://www.nytimes.com/`, including the headline, print headline (if any), url, article id, when the article was first posted online, section, and a subset of the article text (abstract and first paragraph). For each article currently on the home page, the scraper uses the article id to get article metadata (described later) from the official NYT API, `https://developer.nytimes.com/apis`. All of this information is then stored in a database. Note that we can track articles over time using the article id, and we can use this to detect headline changes. The scraper also randomizes over user-agents and does not send cookie headers, which will be essential for detecting A/B tests (most A/B tests use some combination over these factors to assign which group a user is in, and hence which headline he receives).

4.1.1 Defining A/B Tests

The NYT discusses how they conduct A/B tests (our summary is based on Symonds, 2017; Bulik, 2016; Locke and More, 2021). The claim is that various headline writers/editors devise an alternate headline when the article is under-performing relative to expectations. Readers are grouped into equal size groups, with half seeing the original headline and the other half the new alternative (no other aspect of the article is changed). This test goes on for thirty minutes, and the one that attracts the most user interest will then be selected for all users. They have also made mistakes in assigning users to groups in some of their trials. In the empirical analysis we will be evaluate the accuracy of these points.

Two challenges emerge when we attempt to detect these trials.[5] The first is that we observe a random sample of headlines during an A/B test.[6] This means we may not detect the test until part way through the experiment period as we might observe few headline changes. In some cases we may fail to detect an A/B test. For example, we may simply observe the original headline ($AA \ldots A$) even though the NYT server is showing an alternate (B) at times to other users.

The second is that we analyze the data in short time chunks, and these observation periods may not perfectly align with the newspaper's testing period (for example, the test starts on the tenth scrape rather than the first).[7]

Section A in the Appendix models these two issues. We show they have a negligible impact on our detection of A/B tests, in large part because of the high frequency of our data scrapes. With the Appendix results as guidance, we define our observation period as one hour which should exceed the A/B test period (as per the references at the start of this section a period is a half hour). We define an article to have an A/B test when we observe more than one headline change within an hour. As a robustness check, we also defined an A/B test as observing to more than two, three, or four headline changes

[5]We discuss these challenges in more detail in Section A in the Appendix.

[6]The issues in the remainder of this section occur because our data come from being experimental subjects. Most analyses of A/B tests are based on data from the experimenter and perfectly observe all data. Similarly, in our data immediate headline changes are perfectly observed.

[7]The length of the observation period may also differ from the A/B test. We presume the observation period is at least as long as the A/B test period, which means the test can be in no more than two observation periods.

within an hour. The results in our estimations are roughly similar.

4.1.2 Textual Analysis

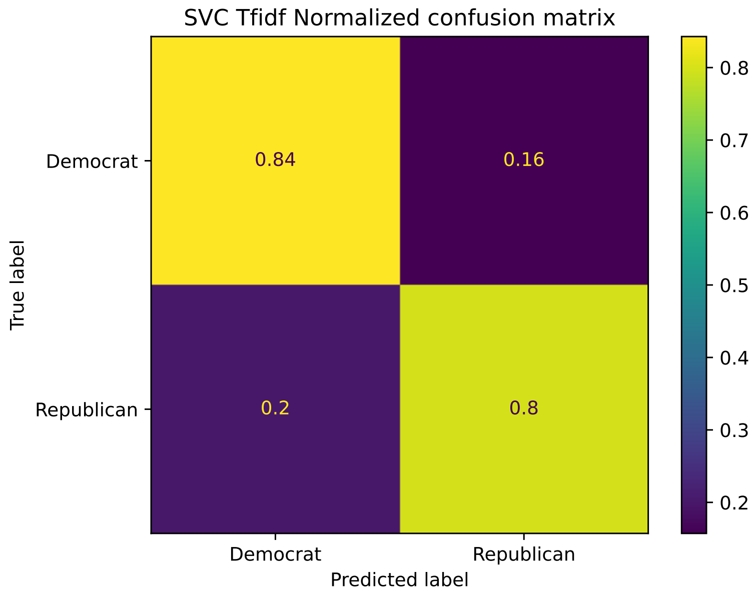
We use textual analysis to construct two measures for the headlines and abstracts (and tweets in later sections) of an article. The first measure is a sentiment score of the text. Because of the lack of training data (headlines or abstracts with labels of their sentiment intensity), we choose to use a dictionary-based approach to implement the sentiment analysis. In particular, we use the VADER sentiment analysis tool (Hutto and Gilbert, 2014) of the Natural Language Toolkit (NLTK) in Python (Bird, Klein, and Loper, 2009) to implement sentiment analysis for the headlines, and abstracts of each article in the sample. As Hutto and Gilbert (2014) shows, VADER's model incorporates syntax and punctuation rules, and is validated with human coding, making its sentence prediction 55–96% accurate, which is on par with Stanford Sentiment Treebank, a method that incorporates a more complex computational algorithm. Also, VADER includes the word banks of established tools like LIWC, ANEW, and GI, as well as special characters such as emoticons and cultural acronyms (e.g., LOL), which makes it particularly suited for social media jargon.[8] In addition to social media text, Hutto and Gilbert (2014) also shows that VADER performs well across several domain contexts including NYT editorials, movie reviews, and product reviews.

The second measure is a measure of political slant. We follow Boxell and Conway (2022) to adopt a machine learning approach to construct our measure of political slant. To construct our training and testing data, we first obtain the Twitter accounts' information of approximately 2,100 politicians from ProPublica, an investigative journalism website.

To apply machine learning, we construct vectorized representations of the tweets. We first apply the standard pre-processing steps to all tweets including lemmatization, punctuation, removals of hashtags, URLs, stop-words, and non-English words. In our application, we use bigrams, which are two-word combinations, as the unit of words and phrases, which yields a better prediction than using unigrams (i.e. single words).To account for the importance of each word, we apply the Term Frequency–Inverse Document Freqency (TFIDF) model, which is the product of the frequency of a word (term frequency, TF) and the importance of the word (inverse document frequency, IDF), in the vectorization

[8]We also use VADER to obtain sentiment scores for tweets which will be discussed in Section 4.2.

Figure 1: Confusion Matrix of Machine Learning Prediction



of tweets. We then use the Linear Support Vector Classification (Linear SVC) model in the scikit-learn package, a Python module, as our machine learning model.

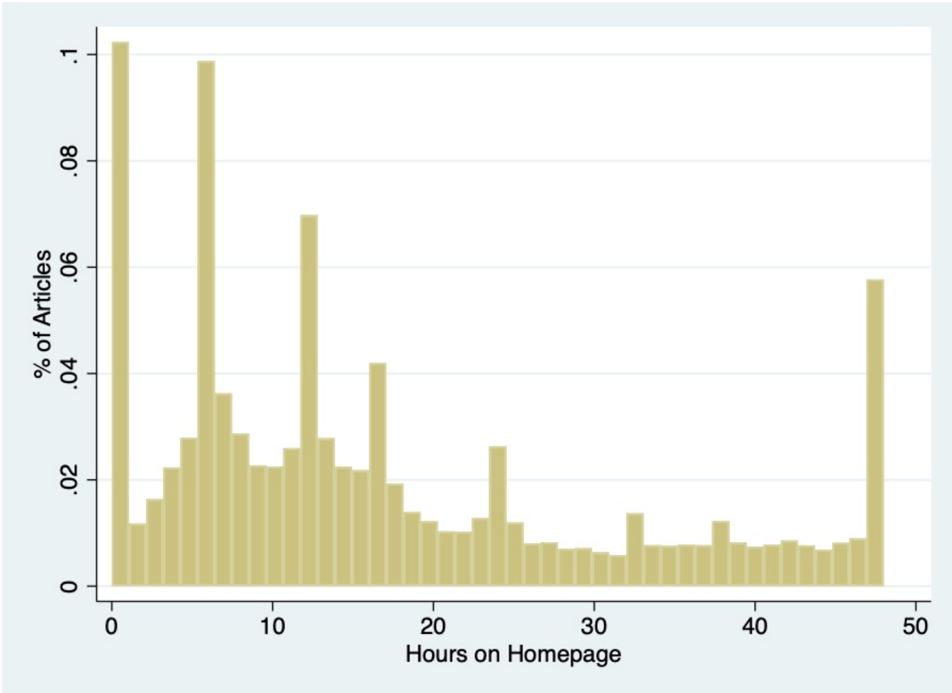
We test the model using the training dataset, which is approximately 1.4 million tweets. Overall, the model achieves 82% accuracy on the leave-out validation dataset. In addition to this, we also construct the confusion matrix, which shows the the percentage of correct and incorrect classifications based on each class (Democrat or Republican), to evaluate the goodness of fit of the model. Figure 1 shows that our model can predict a Democrat's tweet correctly 84% of the time, with only 16% chance being wrongly labeled as a Republican's tweet. It can also predict a Republican's tweet correctly 80% of the time, with 20% chance being wrongly labeled as a Democrat's tweet.

We apply the model to "predict" whether the text in our main dataset is written by a Democrat politician. The confidence of the prediction (in terms of probability of it being written by a Democrat) is our measure of slant.

4.1.3 Summary Statistics

Contrary to the print edition, articles on the digital home page do not have to stay on the front page for exactly one day. Indeed, there are significant variations in terms of articles' length of survival on NYT home page. Figure 2 shows the histogram of the number of

Figure 2: NYT Articles' Length of Survival on Homepage



hours that an NYT article in the sample stays on the home page. There are several things to note. First, an average article stays on the home page for less than a day (16 hours) with a median of survival of 12 hours. Interestingly, slightly more than 10% of all articles stay on the home page for less than an hour. Given that the NYT frontpage is a precious digital real estate, it is reasonable that an underperforming article would be replaced quickly with another that can potentially attract more eyeballs.

Second, a significant portion of articles were pulled from the home page at specific hours. For instance, almost 10% of articles were withdrawn at the 6th hour after they were first published on the home page; approximately 10% of articles were withdrawn at the 11th and 12th hour combined.

Third, less than 6% of the articles stay more than 48 hours. In our analysis, we drop all the observations beyond 48 hours since the article first appears on the home page.

Table 1 reports the summary statistics of the NYT articles by their headline status. Among the 35,164 unique NYT articles in our current sample, the majority (27,523) had never switched their headlines. Among the rest of these articles (7,641), half of them (3,692, 10% of all articles) were immediate headline changes, and half of these articles had conducted an A/B tests on their headlines (11% of all articles).

There are significant variations across different types of articles. NYT GraphQL API metadata divide its articles into three main "tones": "news", "opinion", and ''feature".[9] [10] "News" articles tend to observe more headline changes than articles of other tones. In the overall sample, more than half (55%) are "news", 11% are "opinion", and 34% are "feature" articles. However, the proportion of "news" articles in the sample of articles with headline changes is significantly higher at between 68% and 73%, followed by "opinion" articles (between 16% and 20%), while the proportions of "feature" articles are significantly lower. A "news" article might require an update of its headline when new information regarding the news emerge (such as an outcome of an election), which explains the higher proportion of "news" articles among the immediate switch group. However, the higher proportion of "news" articles among articles that had undergone an A/B test might imply the effectiveness of A/B tests in drawing eyeballs for "news" articles.

While Table 1 does not report these, our data show that articles with headline changes are more popular: For articles which had never switched their headlines, a significantly lower proportion of them had ever hit the top 20 (most viewed) list (between 5% to 10%) compared with articles which had either done a immediate change or undergone an A/B test (on average around 20%). Also, almost half (48.2%) of articles in the sample are in four sections: "US" (18.4%), "world" (11.6%), "opinion" (10.1%), and "arts" (8.2%). Consistent with the tabulation across "tones", proportions of articles in "news"-related sections, especially the "US" section, are significantly higher in the samples of articles with headline changes, which indicates that these articles are more likely to see headline changes of either types.

[9]According to Granato (2002), "[n]ews features are intended primarily to inform, but they frequently also entertain because they utilize human-interest feature writing techniques...Examples are background stories about the new Ministers in the Premier's Cabinet, a counseller's work at a rape crisis centre, a new product whose backers claim will help you lose weight, the most popular automobile models amng car thieves..." This is essentially a type of soft news which could potentially include promoted articles.

[10]A very small number of articles in our sample (0.3% of total) do not have a tone label.

Table 1: Summary Statistics for NYT Articles by Headline Status

	A/B Test	Immediate	No Change	Total
Hours on Homepage	25.13	19.80	15.34	16.90
	(14.37)	(12.80)	(14.00)	(14.29)
sentiment (headline)	-0.0760	-0.0919	-0.0164	-0.0310
	(0.359)	(0.346)	(0.355)	(0.356)
sentiment (abstract)	-0.0547	-0.0560	0.0247	0.00732
	(0.456)	(0.461)	(0.446)	(0.450)
pro-Dem. slant (headline)	0.622	0.626	0.626	0.626
	(0.264)	(0.263)	(0.267)	(0.266)
pro-Dem. slant (abstract)	0.597	0.604	0.607	0.606
	(0.221)	(0.215)	(0.210)	(0.212)
Tone (NEWS)	0.679	0.725	0.510	0.552
	(0.467)	(0.446)	(0.500)	(0.497)
Tone (OPINION)	0.199	0.155	0.0931	0.111
	(0.399)	(0.362)	(0.291)	(0.315)
Tone (FEATURE)	0.123	0.119	0.396	0.336
	(0.328)	(0.324)	(0.489)	(0.472)
N	3949	3692	27523	35164

Mean coefficients; SD in parentheses

Table 1 also reports the sentiment scores of the headlines and abstracts of articles by their headline status. The VADER sentiment scores ranges from -1 (most negative sentiment) to 1 (most positive sentiment). As Table 1 shows, the headlines in the overall sample are, on average, slightly negative (-0.0310) while the abstracts are, on average, slightly positive (0.00732). There are significant differences in both the headlines' and abstracts' sentiments across different types of articles. In particular, articles which had never switched their headlines are, on average, more positive in both their headlines and abstracts.

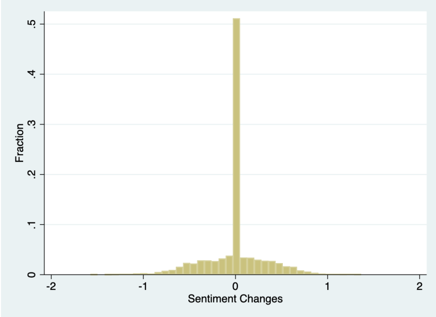
Not surprisingly, the NYT articles in our data contain more pro-Democrat political slant than pro-Republican political slants. The average slant scores of both the headlines and the abstracts of the articles in our sample are approximately 0.6. In other words, our machine learning algorithm predicts, on average, the headlines and abstracts were written by Democrats politicians with a 60% probability. While the headlines' slant were slightly more pro-Democrats than the abstracts', there are not significant variations in the political slant across articles with different headline status.

Among the headlines that were changed either through an immediate change or an

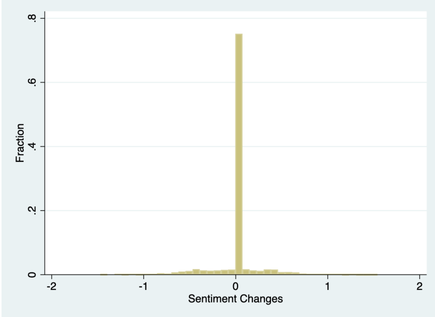
A/B test, a significant portion did not experience any changes in their sentiment and political slant scores. Figure 3 shows that histograms of changes in sentiment scores and political slant scores of headlines which went through either an immediate headline change or undergone an A/B test.

Figure 3: Histograms of Sentiment and Slant Changes in Headline Changes

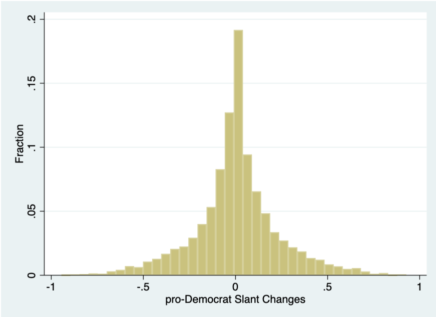
(a)Immediate Change: Sentiment Score



(b) A/B Tests: Sentiment Score



(c) Immediate Change: Slant Score



(d) A/B Tests: Slant Score

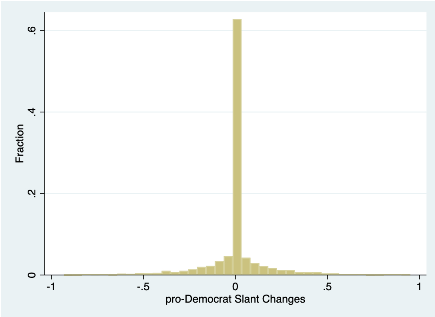


Figure 3a shows that almost 50% of the headlines which went through an immediate headline change did not observe any changes in their sentiment scores, while half of the rest had a more negative headline and another half had a more positive headline after the change. On the other hand, many of the headlines became more polarized politically after an immediate headline changes. Figure 3c shows that more than 90% of the headlines became either more pro-Democrat (49% of total) or more pro-Republican (48% of total) after an immediate headline change.

Figures 3b and 3d show that even more headlines changes did not change the sentiment (73% of total) and political slant (56% of total) scores after going through an A/B test. However, as in the case of immediate headline changes, half of the rest of the headlines saw an increase in sentiment or political slant scores and the remaining half experienced a decrease in these scores.

4.2 Twitter API

Ideally, we would measure the popularity of articles and effectiveness of A/B testing with the actual views and share data from NYT. Unfortunately, we do not have access to it. Instead, we use the number of tweets that link an article's URL as our measure of its popularity. We provide evidence below that twitter users who tweet NYT stories are comparable to the broader NYT readership.

News plays a prominent role on Twitter. In a Pew Research Center study that surveyed 2,548 Twitter users in 2021, Anderson, Rainie, and Nolan (2021) reveal that 23% of Americans use Twitter, and roughly seven-in-ten US Twitter users (69%) say they get news on the site. While only 27% of Americans report to have at least some trust in the infromation they find on social media in general, two-thirds of Twitter news consumers say they have at least some trust in it. Twitter users' reliance on Twitter as a news source has also grown over time. In the same study, the authors report that "fully 70% of Twitter news consumers say they have used Twitter to follow live news events, up from 59% who said this in 2015."

In January 2021, Twitter opened its full tweet archive to academic researchers. We use the Twitter API to extract all tweets containing the URL links of all the NYT articles in our sample. For each tweet, we extract the tweet content, the date and time of the tweet, the number of likes, the number of retweets, and other information of the authors such as the name, the Twitter id, and their self-reported locations. We also construct a sentiment scores for each tweet in the same way as the headlines and abstracts. We end up with a database of over 20 million tweets (including retweets) sent by close to 4 million distinct users.

We use the self-identified user location to determine country, US State, and US Cities. The location field is a free-form string, and N=1,120,911 of the 1,680,259 observations are non-empty. Parsing is challenging, since the location field is sometimes well-formatted ("Brooklyn, NY"; "Chicago metro area"; "Philly"; "Toronto"; "Abuja, Nigeria"; "ÃT: 39.991727,-75.41009"), sometimes has only some of the three geographies ("Indiana, USA"; "Yeehaw Texas"; "America"), and other times is gibberish or non-determinant ("somewhere"; "Under the Contrails"; "daft punk era"; "Global"). For countries, we match this to a list of 241 country names (including variants), their capitals and cities whose metro-wide population in 2018 exceeds 5m people, and large regions (such as New England for the US or Wales for UK). For US States, we match to a list of state names, official and unofficial abbreviations. For US cities, we matched to a list of cities (and abbreviations and variants such as LA, NOLA, KC) with at least 100k population in 2020 (in cases where the same city name appears in multiple locations, like Richmond, Rochester or Springfield, we match to the larger one if there is no additional information), and leading neighborhoods (in New York we use each borough as well as a few dozen neighborhoods like Astoria or Hell's Kitchen). Finally we reverse geocode all cases which could be identified as longitude-latitude pairs (as in the example at the top of the paragraph) using the commercial site OpenCage. After this we spot checked the output, focusing on potential name conflicts (Athens Georgia vs Athens Greece). Finally we used city matches to fill-in missing values for state and country, and state matches to fill-in missing values for country.[11]

In total we match 809,625 users to countries (72% of non-missing rows), 470,907 users to US states (510,282 users are matched to the US), and 279,651 to a US city. Note that in the analysis below we will refer to percent matches in each of these categories, so for example there will be a higher percent of city-matches for New York city than the percent of state-matches for New York state.

4.2.1 Geographic Distribution

The Twitter location reflects the reach of our data.It is also comparable to NYT consumption, which is some evidence that that Twitter activity reflects NYT viewership. Figure 4 shows the distribution by country. This is quite concentrated with the USA alone accounting for almost two-thirds of users, and the top twenty countries include ninety percent of users. Still there are significant number of users on all continents, with Brazil, Colombia, Myanmar, Thailand, Nigeria and South Africa in the top twenty.

We can also compare this to the geographic distribution from SimilarWeb, `https://pro.similarweb.com/`, a publicly traded company which tracks website audiences by following millions of online users and other sources. Our listing of top twenty countries

[11]In matching the various text strings, we first matched using the listed capitalization and then repeated the match using all lower cases (so "New York" would match in the first round and "nEw yorK" would match in the second). For short matches like state abbreviations we only matched using the accepted cases ("Gary, IN" would match to Indiana but "in" would not).

overlaps with SimilarWeb's listing for `www.nytimes.com` (except ours includes Mynamar and theirs includes China), with the US share also being within ten percent from the two sources. This is suggestive evidence that our Twitter users are representative of the broader NYT audience.

Figure 4: Distribution Twitter Users: Countries
Darker shades indicate more users. Top countries: USA (63.0%), UK (5.8%), Canada (3.8%), India (3.0%), France (1.5%).

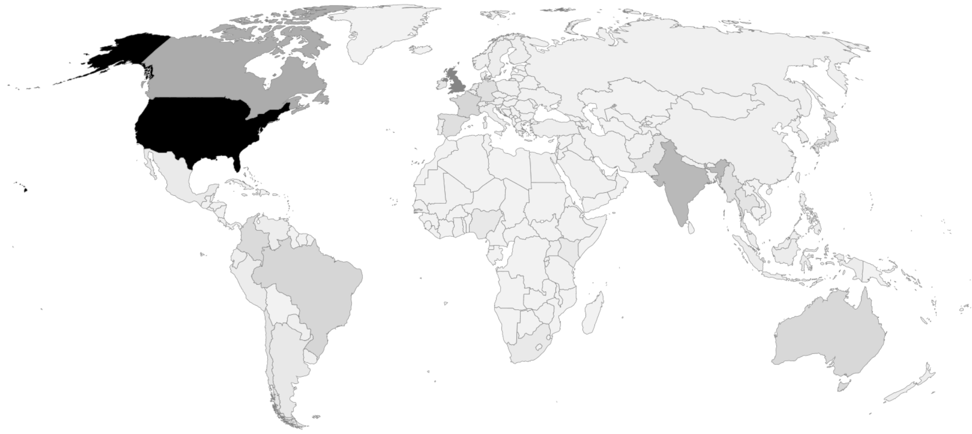


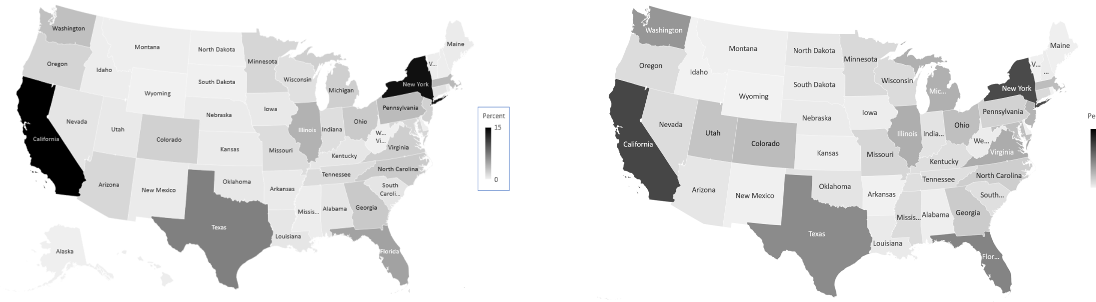
Figure 5a shows twitter users location across US states. This largely follows the population distribution though there are a very large number of users in the District of Columbia.

As with the national geography we can compare to state-level data on NYT readership. Our first source is the National Consumer Survey in MRI-Simmons, `https://www.mrisimmons.com/`. This is a nationally and state representative survey of 25,000 individuals (though it excludes Hawaii and Alaska). Using the 2020 wave, we generate the share of US viewers by state of number of visits to `nytimes.com` over the last thirty days. Figure 5 shows the distribution is quite similar to the Twitter data, and using a Wilcoxon signed-rank test we cannot reject at 5% significance that the two distributions are identical.

Figure 5: Distribution Users: US States (Darker shades indicate more users)

(a) Twitter Top states: California (15.3%), New York (14.0%), Texas (7.1%), Florida (4.9%), District of Columbia (4.3%)

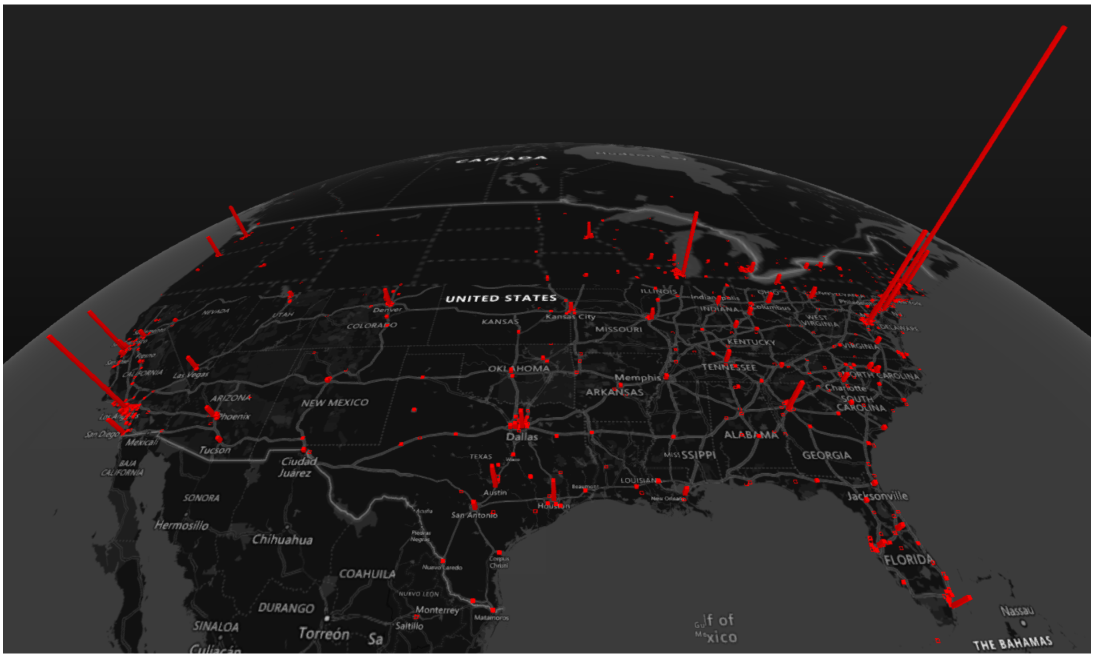
(b) NYT viewers
MRI-Simmons, number visits to `nytimes.com` over last thirty days



Second, we can compare to NYT Sunday print circulation from 2020. The Twitter top twenty states overlaps with the print circulation, with the main exception being that our data have less of a Northeastern concentration (New York and New Jersey have a smaller share and Texas a larger share). Given that print newspapers are more popular near its headquarters, this along with the MRI-Simmons data are indirect evidence that our Twitter data is representative of NYT readership.

Figure 6 has the distribution across US cities. Again the distribution largely follows population though some smaller cities have a disproportionate number of users (District of Columbia, San Francisco, Boston, Atlanta, Seattle, Portland).

Figure 6: Distribution Twitter Users: US Cities
Spike size indicate more users. Top cities: New York (19.2%), District of Columbia (7.3%), Los Angeles (6.5%), Chicago (4.6%), San Francisco (3.5%)



4.2.2 Summary Statistics

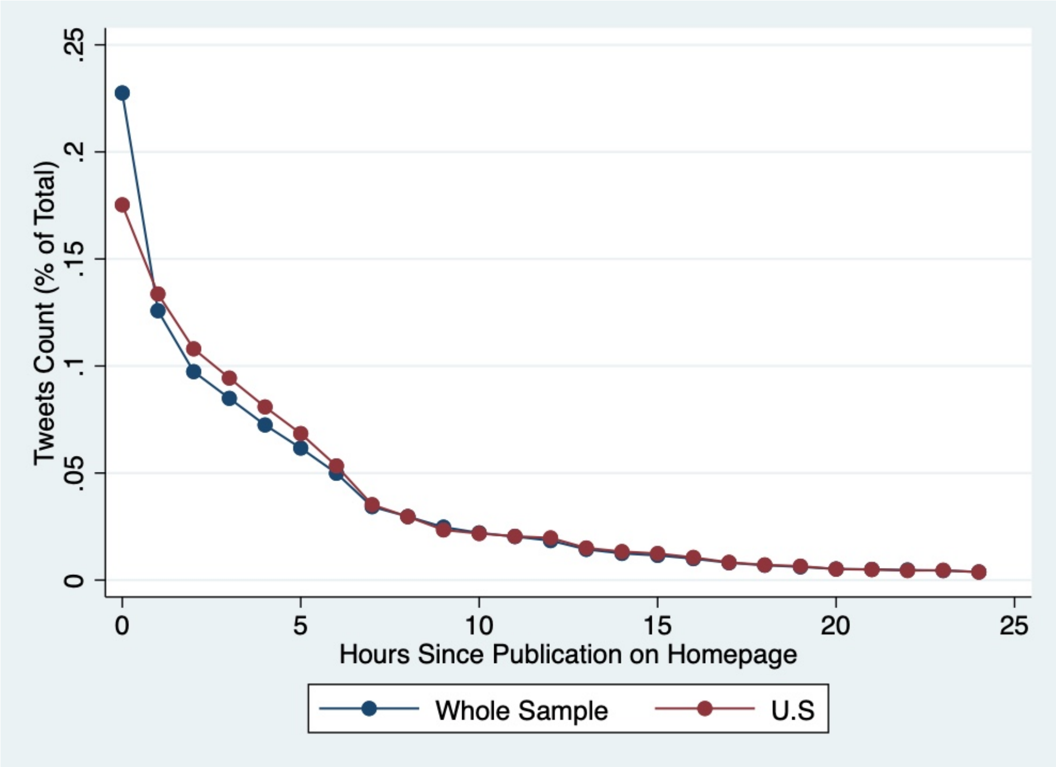
News decay fast in the digital era. Figure 7 shows the average tweets count (as a percentage of total tweets count) of an article's first 24 hours on the NYT home page conditional on that the article has at least one tweet share.[12] On average, an article on the NYT home page generates approximately 23% of its entire tweets count it is first hour on the home page. The tweets count drop by half (to 13% of total) in the next hour, and continues to drop in the hours since. On average, the tweets count at the 12th hour is already less than 2% of its total, and becomes almost zero at the 24th hour mark.

Figure 7 suggests that articles' popularity seems to decay at a slightly slower rate in the US market than in the foreign market. On average, an article generates 18% of its entire US-based tweets count in its first hour, which is 5% lower than that of the overall sample. However, the overall decay pattern between the US sample and the whole sample are similar otherwise.

Table 2 reports the summary statistics of the tweet counts (total, positive, negative,

[12]Approximately 10% and 15% of the articles in our sample have no tweet share and no US-based tweet share at all, respectively.

Figure 7: Decay Pattern of NYT Articles' Tweets Count (Conditional on Tweets Count > 0)



and pro-Democrat) by different groups of articles.[13] Similar to the pattern shown in Table 1, articles which never had headline switch were less popular. Their total tweets count (223) and tweets count per hour (14) are significantly fewer than articles which had an immediate headline change (on average with 471 tweets in total and 26 tweets per hour, and articles which had undergone an A/B test (on average with 568 tweets in total and 24 tweets per hour).

The tweets sentiment of articles which never had headline change are, on average, less negative than the ones that had any type of headline change. While the proportions of positive tweets are the same across different types of articles (approximately 8%), the proportion of negative tweets is significantly lower among the former (5%) than the latter (10%).

Unsurprisingly, these tweets, which mention an NYT article, tend to be pro-Democrats.

[13]In addition to a composite scores, the VADER sentiment analysis also reports the ratio of a set of string (a tweet in our case) that is positive, neutral, and negative. We use the sum of these percentages to construct the positive and negative tweets count per hour. Similarly, our proxy of slant described in Section 4.1.2 measures the likelihood that a tweet is tweeted out by a Democrat, we also use the sum of this likelihood to construct the pro-Democrat tweets count per hour.

On average, approximately 60% of tweets are pro-Democrats. Interestingly, the tweets' slant do not vary significantly across article's types.

Again, we see a similar patter in the US sample.

Table 2: Summary Statistics for NYT Articles' Tweets by Headline Status

	A/B Test	Immediate	No Change	Total
Tweets Count	567.5	471.1	224.9	295.8
(Total)	(1520.1)	(1234.4)	(966.4)	(1090.5)
Tweets Count	24.34	26.29	14.28	16.91
(per hour)	(57.89)	(64.11)	(50.99)	(53.76)
Tweets Count	2.023	2.112	1.245	1.442
(Pos, per hour)	(5.504)	(5.673)	(5.809)	(5.767)
Tweets Count	2.528	2.786	1.335	1.651
(Neg, per hour)	(8.421)	(11.54)	(6.767)	(7.703)
Tweets Count	14.53	16.12	8.775	10.34
(Dem., per hour)	(34.40)	(43.63)	(34.76)	(35.96)

Mean coefficients; SD in parentheses

4.3 Ground News Blindspotter

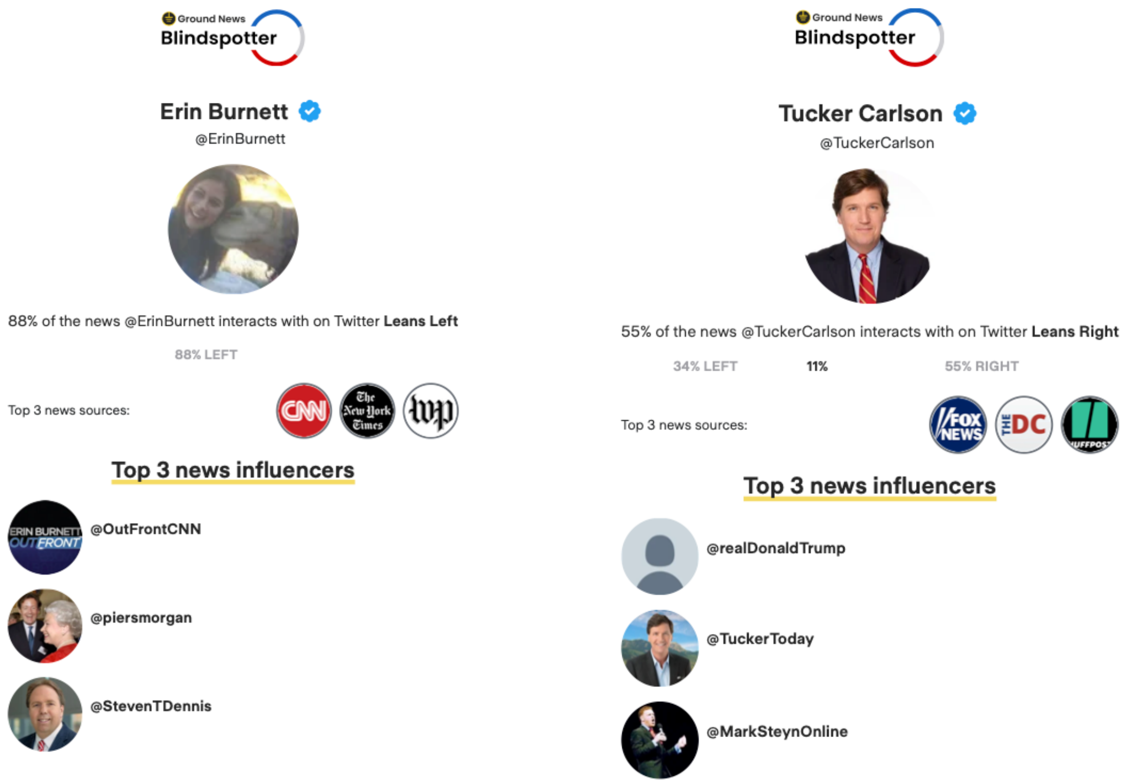
We obtain a measure of each twitter account's partisanship by using a dataset provided by Ground News Blindspotter. Ground News is"a news comparison platform that was created to improve our broken media ecosystem, by abstaining from manipulative algorithms and giving you more control over the news you read". It created the Blindspotter to help people easily visualize his/her news filter bubble.[14]

To measure a Twitter user's partisanship, the Blindspotter tool parses through an account's last 3200 tweets and records interactions (which include tweets, retweets, replies, and likes) with news content that have a bias rating.[15] Its database currently contains

[14]See `https://ground.news/blindspotter/twitter`.

[15]The Blindspotter has its limitations in measuring a Twitter user's partisanship. As it points out in its website: "The data generated by the tool should not be interpreted as exactly representative of someone's political beliefs. An individual's news diet may inform their political beliefs, but it is not the only determinant. Someone who expresses partisan views may have a balanced Twitter news diet and hold those views for different reasons." Other limitations of Blindspotter include 1) its tool cannot decipher if an interaction is positive or negative; 2) the results of the tool may be less meaningful for accounts largely run by assistants/staffers; 3) its tool only records interactions with news sources that have a bias rating, so smaller local publications may not be included as a "news interaction".

Figure 8: News Diet of Erin Burnett and Tucker Carlson



2,063 news sources with bias ratings, which includes 878 left-leaning sources, 571 right-leaning sources and 614 that are centrist.[16] It places different weights on interactions with news content to adjust for their importance in revealing one's partisanship. In particular, one " like" is given a weight of 1, one "tweet" a weight of 2, one "retweet" a weight of 1.5, one "quote" a weight of 1.45, and one reply a weight of 1.85.

Figure 8 is an example of how the data look like on Blindspotter. The news diet of the CNN anchor, Erin Burnett (left), is comprised of 88% left-leaning news, and her three top sources of news are CNN, NYT, and Washington Post. On the other hand, the news diet of Tucker Carlson (right), the Fox News anchor, is mainly comprised of right-leaning news (55%) with only 34% of his news diet being right-leaning.

Table 3 reports the news diet of the Twitter accounts who shared NYT articles in our sample. As expected, the average news diet of these Twitter accounts comprised of a majority of left leaning news. On average, 68% of the news these account share are left-leaning, while only 8% of the news are right-leaning, and 24% are centrist. Articles

[16] NYT is classified as a left-leaning source by all three of its bias ratings source: allsides, mediabiasfactcheck and adfontesmedia.

with headline changes tend to attract more polarized Twitter accounts than articles without headline change. The proportion of centrist news in the former is 21%, while the corresponding number of the latter is 25%.[17]

On average, almost 18% of the news diet of the Twitter accounts in our sample are NYT, with not much variation across different types of articles. The only exception is that articles which had an A/B test tend to attract Twitter accounts with a lower percentage of NYT in their news diet. The pattern is similar in the US sample, though they tend to have more NYT in their news diet.

Table 3: Summary Statistics for Twitter Accounts' News Diet by Headline Status

	A/B Test	Immediate	No Change	Total
Blindspotter	0.702	0.701	0.679	0.684
(Left, Ratio)	(0.0791)	(0.0806)	(0.129)	(0.120)
Blindspotter	0.0922	0.0905	0.0750	0.0785
(Right, Ratio)	(0.0670)	(0.0646)	(0.0642)	(0.0648)
Blindspotter	0.164	0.183	0.183	0.182
(NYT, Ratio)	(0.0754)	(0.0894)	(0.111)	(0.106)

Mean coefficients; SD in parentheses

5 Empirical Strategy and Results

We attempt to do two things in this section. First, we document the types of articles which have a higher likelihood to have a headline change (both immediate change and A/B test). Second, we investigate the impact of a headline change on several metrics of the article's performance.

5.1 Decision on Headline Switch and A/B Testing

5.1.1 Which Types of Articles Get Headline Changes?

To understand which types of articles are more likely to get headline changes, we run the following simple multinomial logit regression to find the correlation between whether an

[17]We see a similar pattern in the US sample. The main difference is that US Twitter accounts are even more left-leaning, with their average news diet comprising of 74% of left-leaning news, and only 6% of right-leaning news.

article got a headline switch or A/B testing. Let us first define $k \in [NoChange, Immediate, AB]$ be the three types of article (No headline change, immediate headline change, A/B test) in our data. The likelihood of article i being type k is therefore:

$$\Pr Y_{ik} = \frac{\beta_{0k} + \beta_{1k} X_i^{info} + \beta_{2k} X_i^{twitter} + \beta_{3k} X_i^{blindspotter}}{\sum_{j=1}^{3} \beta_0 + \beta_{1j} X_i^{info} + \beta_{2j} X_i^{twitter} + \beta_{3j} X_i^{blindspotter}}, \tag{1}$$

X^{info} is a vector of article-level variables such as the number of hours between original publication and appearance on home page, article's tone dummies, section dummies e.t.c. reported in Table 1, $X^{twitter}$ is a vector of article-level variables related to the article's tweets such as its first hour tweets count and first hour tweets sentiment reported in Table 2, $X^{blindspotter}$ is a vector of article-level variables related to the average news diet of the twitter accounts which share the article, such as the ratio of left- and right-leaning news in these accounts' news diet, reported in Table 3.

Table 4 reports the results.[18] The upper panel reports the coefficients for immediate headline change, while the lower panel reports the coefficients for A/B test. Several things are worth pointing out. First, articles which are seemingly intrinsically more popular have a higher likelihood to have a headline change and an A/B test. For instance, articles which take a shorter time to make it to the home page are more likely to have headline changes or A/B testings. On average, an article that make to the home page 10 hours earlier are approximately 1 percentage points and 3 percentage point more likely to have a headline change and undergo an A/B test, respectively. This is reasonable given that intrinsically more popular articles tend to be picked up by the editor to appear on the home page earlier.

Also, articles which are more popular on Twitter within the first hour (another indication that the articles are intrinsically more popular) are also more likely to have a headline swtich or undergo an A/B test. On average, a one standard deviation increase in the first hour tweets count is correlated with 0.4 percentage points and 0.5 percentage points increases in the likelihood of a headline change and an A/B test, respectively.

[18]The pattern of the results are broadly similar using the US sample.

Table 4: Which Articles Get Headline Swtich or A/B Testing?

	(1)	(2)	(3)
Immediate Change			
From Publication	-0.00111***	-0.000920***	-0.000872***
to Homepage (hrs)	(-10.27)	(-5.77)	(-5.36)
Tweets Count	0.0000698***	0.0000758***	0.0000666***
(First Hour)	(5.15)	(5.04)	(4.43)
Headline Sentiment		-0.00708	-0.00637
(First Hour)		(-1.06)	(-0.95)
Tweets Sentiment		-0.0158	-0.0164
(First Hour)		(-1.75)	(-1.81)
Sentiment Difference		0.00550	0.00506
(Headline - Abstract)		(1.23)	(1.13)
Blindspotter (Left)			0.000202*
(First Hour)			(2.19)
Blinespotter (Right)			0.0000615
(First Hour)			(0.22)
A/B Test			
From Publication	-0.00270***	-0.00341***	-0.00314***
to Homepage (hrs)	(-16.24)	(-13.01)	(-11.96)
Tweets Count	0.000106***	0.000116***	0.0000995***
(First Hour)	(8.21)	(8.28)	(7.17)
Headline Sentiment		-0.00725	-0.00449
(First Hour)		(-1.07)	(-0.67)
Tweets Sentiment		-0.00763	-0.00730
(First Hour)		(-0.82)	(-0.80)
Sentiment Difference		0.00594	0.00465
(Headline - Abstract)		(1.31)	(1.03)
Blindspotter (Left)			0.000620***
(First Hour)			(6.33)
Blinespotter (Right)			0.00167***
(First Hour)			(6.43)
Section FE	Yes	Yes	Yes
Hour FE	Yes	Yes	Yes
Observations	35164	27934	27866

Coefficients are marginal effects. t statistics in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Negative (in terms of sentiment scores) articles tend to see more headline changes. On average, more negative headlines and more negative sentiments on Twitter that articles generate tend to lead to more immediate headline changes and A/B tests, though the correlations are statistically insignificant. The correlations (and their statistical significance) are also weaker when the Blindspotter controls are included. However, a more positive headline (compared to abstract) seem to have a positive correlation with an immediate headline change and an A/B test. This might suggest that an immediate change or an A/B test is one way to "correct" for the "misalignment" of an article's headline sentiment relative to its content.

The news diet of Twitter accounts that share the NYT articles are (statistically) significantly correlated with the immediate headline changes and A/B testing decisions. In particular, articles which attract more partisan Twitter accounts tend to have a higher likelihood of an immediate headline change and an A/B test, though the magnitude is not large.

5.1.2 Social Media Pressure on Headline Changes

As Cagé, Hervé, and Mazoyer (2022) show empirically using 2 billion tweets in France, popularity of a story in social media can lead to increases in the coverage of the same story by mainstream media. Similarly, it is not uncommon for media to adjust their content (and to change their headlines) in the wake of pressure from the readers. On August 3, 2019, a mass shooting occurred at a Walmart store in El Paso, Texas, in which 23 people were killed and another 23 people were injured. Two days later, President Trump, in remarks at the White House, condemned white supremacy and called for the death penalty for mass murderers and domestic terrorists.[19]

The NYT first carried a report on the speech with the headline: "Trump Urges Unity Versus Racism". While the content of the article was critical of Trump's speech, the headline drew criticisms from politicians and influencer from the left side of the political spectrum. Alexandria Ocasio-Cortez, the Democrat Congresswoman, claimed it was "a reminder of how white supremacy is aided by—and often relies upon—the cowardice of mainstream institutions".[20] An NYT op-ed writer, Wajahat Ali, said the headline is "terrible".[21] An NYT bestselling author, Chuck Wendig, said "The NYT is so disappointing."[22] Nate Silver, the founder of FiveThirtyEight, said he would have framed

[19]Much of the description of the chain of events is drawn from here: https://www.newslaundry.com/2019/08/09/new-york-times-headline-changed-donald-trump.

[20]See https://twitter.com/AOC/status/1158557383767052293.

[21]See https://twitter.com/WajahatAli/status/1158552240719912962.

[22]See https://twitter.com/ChuckWendig/status/1158549383320887297.

the headline differently.[23]

Not long after that, NYT issued a new headline: "Assailing Hate But Not Guns", which immediately drew the criticism from President Trump himself. In his now-suspended tweet, he slammed NYT for surrendering to pressure. The headline was then changed once again to "Trump Condemns White Supremacy but Stops Short of Major Gun Controls".

In an attempt to explain the multiple headline changes within a short space of time, a deputy managing editor at NYT stated that it was internal complaints that triggered the changes:[24] "We needed to deliver a nuanced message in a very small space under tight deadlines, and unfortunately, our first attempt at that did not hit it right. When a group of top editors received an email with the first edition of the front page last night, we saw the headline, realized that it was not a good one and decided to change it. It's not uncommon for our masthead editors to adjust headlines as we go." However, he also acknowledged that he was aware of the "heated discussion" on Twitter about the headlines: "As this conversation was happening among Times editors, readers began discussing the initial headline on Twitter. They rightly pointed out that the initial headline didn't reflect the story accurately."

In this section, we attempt to use our Twitter data to analyze whether social media pressure affects headline changes at NYT. We construct our social media pressure controls in the following ways. First, we collect the twitter accounts information of two NYT-related groups. The first group is the bylines of the articles in our main data. Among the 35,164 articles in our main data, we are able to find the Twitter accounts of the bylines of approximately 15,000 articles. The second group is the NYT official accounts, such as @nytimes, @nytopinion, and @nytpolitics etc.[25]

Second, we identify two types of tweets that would potentially catch the attention of, and thus put pressure on, the editors at NYT, namely the tweets sent out by accounts which either the bylines or the NYT official accounts follow, and the tweets which mentioned these accounts.

Third, for each type of these tweets, we construct three measures of social pressure (per hour), namely the number of tweets with positive sentiment, with negative sentiment

[23] See `https://twitter.com/NateSilver538/status/1158546596252061696`.

[24] Full statement can be read here: `https://www.nytimes.com/2019/08/06/reader-center/trump-mass-shootings-headline.html?mod=article_inline`.

[25] There are 43 official NYT accounts in total. The list members can be found here: `https://twitter.com/i/lists/830071071407759360/members`.

and with pro-Democrat slant.

We then run the following OLS regressions with article fixed effects:

$$Headline_{it} = \alpha_0 + \alpha_1 Pos_{it} + \alpha_2 Neg_{it} + \alpha_3 Dem_{it} + \alpha_4 X_{it}^{pressure} + \delta_i + \varepsilon_{it}, \tag{2}$$

where $Headline_{it}$ is a dummy variable for either an immediate headline change or an A/B test of article i at hour t, $Pos_{it}, Neg_{it}, Dem_{it}$ are the three measures of social pressure (positive tweets, negative tweets, and pro-Democrat tweets) of article i at $t-1$, $X_{it}^{pressure}$ is a vector of other article-related controls such as the number of hours on homepage at t, the total number of tweets of article i at $t-1$ (which controls for the hour-by-hour popularity of the article), hour-of-the-day fixed effects etc, δ_i is article i fixed effects which control for the intrinsic popularity of the article.

Table 5 reports the results for pressures from different groups on the likelihood of an immediate headline change and an A/B test, respectively. Columns 1 and 2 in the tables report the results from pressures from accounts followed by and mentioned either the bylines or the NYT official accounts on immediate headline changes, while columns 3 and 4 report the results on A/B tests.

Several things are worth noting. First, headline changes are likely to be done early in an article's appearance on NYT homepage. The chances of an immediate headline change and an A/B test decrease by 0.05 percentage points and 0.1 percentage points each hour, respectively.

Table 5: Social Pressure on Headline Changes

	(1) Immedidate (Follow)	(2) Immediate (Mention)	(3) A/B Test (Follow)	(4) A/B Test (Mention)
Hours on Homepage	-0.000544*** (0.0000200)	-0.000554*** (0.0000200)	-0.00101*** (0.0000225)	-0.00102*** (0.0000224)
Social Pressure (pos, Lag Tweets)	0.00170* (0.000957)	-0.0000320 (0.0000912)	-0.00136 (0.00107)	0.000206** (0.000102)
Social Pressure (neg, Lag Tweets)	0.00402*** (0.000827)	0.000000424 (0.0000761)	0.00268*** (0.000929)	0.000208** (0.0000855)
Social Pressure (slant, Lag Tweets)	-0.000199 (0.000243)	0.0000441** (0.0000209)	0.000306 (0.000273)	-0.0000463** (0.0000235)
Observations	448884	448884	448884	448884
Hour FE	Yes	Yes	Yes	Yes
Article FE	Yes	Yes	Yes	Yes
R^2	0.00	0.00	0.01	0.01
F-Stat	63.12	61.76	93.82	93.32

Standard errors in parentheses
* $p < .1$, ** $p < .05$, *** $p < .01$

Second, NYT editors seem to be more aware of tweets from accounts that they (bylines and NYT official accounts) follow on Twitter than from tweets that mentioned them. Indeed, almost none of the pressure variables from tweets that mentioned them is statistically significantly correlated with headline changes.

Among the pressures from accounts that they follow on Twitter, headline changes are more correlated with negative tweets than with positive tweets. For instance, each negative tweet from an account that NYT editors follow is correlated with a 0.4 percentage point increase in the likelihood of an immediate headline change, but a positive tweet from the same account is only correlated with 0.1 percentage point increase in the likelihood. In fact, positive tweets from these accounts are *negatively* correlated with A/B tests: one positive tweet from these accounts is correlated with 0.1 percentage point *decrease* in the likelihood of an A/B test.

Last but not least, pro-Democrat tweets from accounts that they follow on Twitter do not have a statistically significant correlation with immediate headline changes. This, in a sense, corroborates with the anecdote of NYT's reporting of President Trump's address on the El Paso mass shooting in which NYT changed the headline twice, once under the pressure from the left, another under the pressure from President Trump. However, pro-Democrat tweets from these accounts are positively correlated with an A/B test.

Most of the results mentioned in Table 5 are driven by hard news. Table 6 reports the results of social media pressure on articles with different news tone. Neither feature nor opinion articles tend to change headlines under social media pressure. If anything, opinion articles tend to have a lower likelihood to see an immediate headline change when there are more negative tweets. On the other hand, hard news (columns 2 and 5) articles tend to see more immediate headline changes and A/B tests when there are negative tweets about the articles.

Table 6: Social Media Pressure by Tone

	(1) Immediate (Feature)	(2) Immediate (News)	(3) Immediate (Opinion)	(4) A/B Test (Feature)	(5) A/B Test (News)	(6) A/B Test (Opinion)
Hours on Homepage	-0.000196***	-0.000640***	-0.000673***	-0.000586***	-0.00123***	-0.00107***
	(0.0000289)	(0.0000387)	(0.0000285)	(0.0000367)	(0.0000417)	(0.0000338)
Social Pressure (pos, Lag Tweets)	0.00622***	0.00100	0.00190	0.00243	-0.00171	-0.00267
	(0.00198)	(0.00124)	(0.00352)	(0.00252)	(0.00133)	(0.00417)
Social Pressure (neg, Lag Tweets)	0.00293	0.00443***	-0.00923**	0.000377	0.00289***	-0.00213
	(0.00262)	(0.00101)	(0.00391)	(0.00333)	(0.00109)	(0.00464)
Social Pressure (slant, Lag Tweets)	-0.000472	-0.000307	0.00183	0.00124*	0.000141	-0.000508
	(0.000592)	(0.000304)	(0.00122)	(0.000753)	(0.000327)	(0.00144)
Observations	110065	221619	117087	110065	221619	117087
Hour FE	Yes	Yes	Yes	Yes	Yes	Yes
Article FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.00	0.01	0.01	0.00	0.01	0.01
F-Stat	8.98	39.22	32.22	12.81	46.93	46.87

Standard errors in parentheses
* $p < .1$, ** $p < .05$, *** $p < .01$

5.2 Impacts of Immediate Headline Changes and A/B Tests

In this section, we investigate the impact of having an immediate headline change or undergoing an A/B test of headlines on several outcome variables, including the number of tweets, tweets' sentiments, tweets' political slant, and the news diet of Twitter accounts. We present results from two approaches, namely propensity score matching and instrumental variable. Appendix Section B shows headline changes do not displace attention from the other articles not getting a change, a possibility that would complicate the interpretation of the estimates in this section.

5.2.1 Propensity Score Matching

The propensity score matching (PSM) estimator evaluates the effect of headline change (immediate headline change or A/B test) by comparing the outcome variables for articles that received a headline change to those of similar article as a comparison group. Essentially, the PSM average treatment effect (ATE) estimator answers the question: "What is the expected effect on the outcome variable if articles in the population were randomly selected for headline change?" Denote $P(Z) = P(D = 1|Z)$ as the probability for an article to receive headline change ($D = 1$) given its observed characteristics Z.[26] The PSM estimator for the ATE of headline change is:

$$\pi_{ATE}^{PSM} = E_{P(Z)}\{E[Y(1)|D = 1, P(Z)] - E[Y(0)|D = 0, P(Z)]\}, \tag{3}$$

where $Y(1)$ and $Y(0)$ are the outcome variables of articles that did and did not receive headline change, respectively. Overall, our estimator is able to find similar matches for the treated articles. In particular, the baseline covariates are similarly distributed between the treated and untreated groups. Table 7 shows that the standardized mean differences between group means across all baseline covariates are close to zero, the variance ratios also shrink by half.[27]

[26]These controls include article-level controls such as the number of hours on homepage at t, hour-of-the-day fixed effects, year-month fixed effects, section fixed effects, and controls related to the articles' popularity such as the total number of tweets, the number of positive tweets, the number of negative tweets, and the number of pro-Democrat tweets of the article at $t-1$, and the three measures of social pressure in Equation (2).

[27]We do not show the standardized mean differences of the hour-of-the-day dummies, year-month dummies, section dummies and news tone dummies, which are also close to zero, for space constraint reason.

Table 7: Standardized Differences and Variance Ratios of Matching Sample

(a) Immediate Change

	Standardized Dif.		Variance Ratio	
	Raw	Matched	Raw	Matched
Hours on Homepage	-.4634773	-.0010701	.4837879	1.035777
Lag Tweets	.1237964	.0330256	1.691857	.7824414
Lag Tweets (Dem.)	.1129336	.024504	1.779128	.7115604
Lag Tweets (Pos)	.0807221	.0233503	1.536545	.9752235
Lag Tweets (Neg)	.1077392	.0240564	2.133867	.8236484
Social Pressure (Dem., follow)	.1757088	.0501554	2.400239	1.076914
Social Pressure (Pos follow)	.1370034	.0325384	2.55796	1.193522
Social Pressure (Neg follow)	.1642708	.0507898	3.047735	1.167248
Social Pressure (Dem., mention)	.0757405	.0180149	3.219304	1.425075
Social Pressure (Pos, mention)	.0482329	.0166181	2.530351	1.918022
Social Pressure (Neg mention)	.0579077	.0127119	2.443616	1.868017

(b) A/B Tests

	Standardized Dif.		Variance Ratio	
	Raw	Matched	Raw	Matched
Hours on Homepage	-.1505958	.018791	.8047779	.9886839
Lag Tweets	.157153	.0546864	1.842919	.7384401
Lag Tweets (Dem.)	.1411871	.0500207	1.610816	.7162223
Lag Tweets (Pos)	.1052372	.0342705	1.926229	.9447122
Lag Tweets (Neg)	.1408031	.0456477	1.753155	.6530627
Social Pressure (Dem., follow)	.1468637	.0483878	2.221901	1.156355
Social Pressure (Pos follow)	.1079131	.0356821	2.056213	1.281151
Social Pressure (Neg follow)	.1417603	.0371267	3.150273	1.197935
Social Pressure (Dem., mention)	.0948454	.0489012	1.625098	1.085278
Social Pressure (Pos mention)	.0558813	.0255257	2.353138	1.73296
Social Pressure (Neg, mention)	.0846941	.0219974	1.67965	.6876003

Table 8 report the results of our matching estimation for the impact of immediate headline change and A/B tests, respectively.[28]

Table 8: Impact of Headline Changes (PSM)

(a) Immediate Change

	(1) # Tweets (All)	(2) # Tweets (Negative)	(3) # Tweets (Positive)	(4) # Tweets (Dem.)
ATE	6.554***	0.541***	0.558***	4.264***
	(1.516)	(0.159)	(0.158)	(1.125)
Observations	400621	400621	400621	400621
Hour FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes

Standard errors in parentheses
* $p < .1$, ** $p < .05$, *** $p < .01$

(b) A/B Tests

	(1) # Tweets (All)	(2) # Tweets (Negative)	(3) # Tweets (Positive)	(4) # Tweets (Dem.)
ATE	5.419***	0.614***	0.357***	3.270***
	(0.772)	(0.0961)	(0.0826)	(0.482)
Observations	403583	403583	403583	403583
Hour FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes

Standard errors in parentheses
* $p < .1$, ** $p < .05$, *** $p < .01$

Overall, in the hour after a headline change, the article would become more popular. Column 1 in Table 8a shows that, compared to articles in the control group, there would be 6.5 more tweets (a 0.1 standard deviation increase or 38% of the mean) sharing the article the hour after the headline change. Columns 2 and 3 suggest that the sentiment of the additional tweets are similar to the average (roughly less than 10% of the tweets being positive and negative according to Table 2). The same is true on the impact of immediate headline changes on pro-Democrat tweets: approximately 2/3 of the additional tweets are

[28] Table A4 in the Appendix shows that the results are similar (albeit smaller in magnitude) for the average treatment effect on the treated (ATET), where the PSM estimator is defined as: $\pi^{PSM}_{ATET} = E_{P(Z)|D=1}\{E[Y(1)|D=1, P(Z)] - E[Y(0)|D=0, P(Z)]\}$.

pro-Democrat, which is in line with the average.

The impact of A/B headline tests are similar. Column 1 in Table 8b shows that tweets share would increase by 5.4 (a 0.08 standard deviation increase or 32% of the mean) compared to articles in the control group. Similarly, most of the additional tweets (60%) are pro-Democrat (Column 4), which is again similar to the average. However, A/B tests seem to generate more negative sentiment than immediate changes, with 11% of these additional tweets being negative (Column 2) and only 7% of them being positive (Column 3).

There are several things worth noting. First, the magnitude of the headline change impacts are smaller compared to the IV estimates which will be shown in Section 5.2.2. However, these PSM esatimates are still statistically significant and economically meaningful (in the sense that they are more than 30% of the mean).

A related point is that the impact of headline changes are heterogenous across articles of different popularity. Section C in the Appendix compares the transition matrix of the treatment group and control group, which shows that most of these impact of headline changes concentrated on articles with tweet counts between 1 and 22 tweets.

As we showed in Figure 3, headline changes are heterogenous. For instance, headline could be changed to reflect a more negative sentiment or consist of more pro-Democrat slant, which could have a different impact on the various performance metrics of an article with a more positive or more pro-Republican headline change. To investigate the heterogenous impact of headline changes, we divide the headline changes into three groups ("down" if the score went down after the change, "same" if it remained the same, and "up" if it increased) for each of the headline change type.

Table 9 reports the PSM estimates on tweet shares for each headline change type. Overall, the impact of immediate headline changes do not vary regardless of the changes in sentiment scores (Table 9a) and changes in pro-Democrat political slant (Table 9c). If anything, headlines changes with the same sentiment score or political slant score tend to have slightly bigger impact on tweet shares than otherwise, but the differences are small.

On the other hand, more polarized headlines in sentiment after an A/B test (Columns 1 and 3 in Table 9b) seem to have a bigger impact (more than double) on tweet shares than other A/B tested headline changes that did not result in changes in sentiment. (Column 2) Table 9d shows similar (but less dramatic) results on comparing headline changes in different political slant groups.

To the extent that these responses (in terms of additional tweet shares) reflect readers' taste for sentiment and political slant, these results regarding the impact of A/B test in Tables 9b and 9d suggest that polarization of headlines are demand driven.

Table 9: Heterogenous Impact of Headline Changes on Tweet Shares (PSM)

(a) Immediate Change (Sentiment)

	(1) Down	(2) Same	(3) Up
ATE	3.875**	4.910***	4.860***
	(1.594)	(1.833)	(1.861)
Observations	395490	397639	394082
Hour FE	Yes	Yes	Yes
Month FE	Yes	Yes	Yes

Standard errors in parentheses
* $p < .1$, ** $p < .05$, *** $p < .01$

(b) A/B Tests (Sentiment)

	(1) Down	(2) Same	(3) Up
ATE	8.233***	3.940***	9.222***
	(2.272)	(0.829)	(2.894)
Observations	394078	401276	392212
Hour FE	Yes	Yes	Yes
Month FE	Yes	Yes	Yes

Standard errors in parentheses
* $p < .1$, ** $p < .05$, *** $p < .01$

(c) Immediate Change (pro-Democrat)

	(1) Down	(2) Same	(3) Up
ATE	5.396***	7.774	4.736***
	(2.049)	(5.138)	(1.421)
Observations	397622	355449	397695
Hour FE	Yes	Yes	Yes
Month FE	Yes	Yes	Yes

Standard errors in parentheses
* $p < .1$, ** $p < .05$, *** $p < .01$

(d) A/B Tests (pro-Democrat)

	(1) Down	(2) Same	(3) Up
ATE	6.949***	3.767***	4.120**
	(1.436)	(0.887)	(1.978)
Observations	395931	399835	396969
Hour FE	Yes	Yes	Yes
Month FE	Yes	Yes	Yes

Standard errors in parentheses
* $p < .1$, ** $p < .05$, *** $p < .01$

Headline changes' impact also vary by news tones. Table 10 shows the results by news tone, which suggests that most of the headline changes impact come from news articles.[29] Columns 2 in Table 10a and 10b show that tweet shares of the article would go up by 6.1 and 8.1 the hour after an immediate headline change and A/B test compared to the control group, respectively. Feature articles' tweet share increase by less than 2 the hour after a headline change (Column 1 in both sub-tables). Opinion articles' tweet share increases by more than 4 the hour after an immediate headline change, but do not increase (both statistically and economically) significantly after an A/B test on the headline.

[29]Results are similar from the PSM ATET estimator in Table A5 in the Appendix.

Table 10: Impact of Headline Changes on Tweet Shares by Tone (PSM)

(a) Immediate Change

	(1) Feature	(2) News	(3) Opinion
ATE	1.969***	6.052***	4.431***
	(0.566)	(1.684)	(1.686)
Observations	96766	193703	110152
Hour FE	Yes	Yes	Yes
Month FE	Yes	Yes	Yes

Standard errors in parentheses
* $p < .1$, ** $p < .05$, *** $p < .01$

(b) A/B Tests

	(1) Feature	(2) News	(3) Opinion
ATE	1.766***	8.119***	0.662
	(0.532)	(1.440)	(0.625)
Observations	97369	195301	110913
Hour FE	Yes	Yes	Yes
Month FE	Yes	Yes	Yes

Standard errors in parentheses
* $p < .1$, ** $p < .05$, *** $p < .01$

5.2.2 Instrumental-Variable Approach

On top of using PSM to identify the impact of headline changes, we also employ an instrumental-variable approach. For each type of headline change (immediate headline change or A/B test), we run the following regression:

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \beta_2 Y_{it-1} + \beta_3 Headline_{it-1} + \delta_i + \epsilon_{it}, \tag{4}$$

where Y_{it} is the outcome variable of interest for article i at hour t since its appearance on NYT home page, X_{it} is a vector of variables of article i at hour t which includes the hours on home page (and its square) and hour-of-day dummies, Y_{it-1} is the lag variable of the outcome variable, $Headline_{it-1}$ is a dummy variable which indicates whether article i underwent headline change (immediate headline change or A/B test) at hour $t-1$. We also include article fixed effects (δ_i) to control for unobserved heterogeneity such as intrinsic news-worthiness or popularity at the article level in some specifications.

The empirical challenge in the estimation is the endogeneity of $Headline_{it-1}$. Editors do not exogenously pick an article at a particular hour for headline change. For instance, more popular articles (hence with more tweets count) are more likely to experience a headline change. While we have article fixed effects (δ_i) and the lag variable of the outcome variable (Y_{it-1}) to (partially) control for this factor, there can still be unobserved heterogeneity that is correlated with the editors' decision to make headline changes for an article. In other words, $Headline_{it-1}$ and ϵ_{it} might be correlated.

We use an instrumental-variable (IV) approach to address the potential endogeneity issue. Our proposed instrument is the number of headline changes observed in other articles at the same hour:

$$IV_{it} = \sum_{j \neq i} Headline_{jt}.$$

The intuition of the choice of our instrument is as follow: Time is scarce in the media industry. Editors do not have time to meet every hour to discuss headline changes. Therefore, the economies in scale in discussing headine changes would mean multiple headline changes observed in the same hour. Indeed, in our correspondence with a senior editor of the digital transition team at the NYT, he admitted that "[I]t is not unusual to run simultaneous A/B tests running on different homescreen articles." To the extent that the unobserved heterogeneity of other articles are not correlated with the outcome variable of article i, i.e. ϵ_{jt} and Y_{it} are not correlated, IV_{it} would be a good candidate as an instrumental variable in our specification.

Table 11 reports results from the following OLS regressions:

$$Headline_{it} = \alpha_1 X_{it} + \alpha_2 IV_{it} + \delta_i + \varepsilon_{it}. \tag{5}$$

The first two columns of Table 11 report the results for immediate headline changes, with the first column uses the number of other headline changes in the same section in the same hour as the IV, while the second column uses the number of other headline changes in all sections as the IV. The last two columns report the corresponding results for A/B tests. The results suggest that the number of other headline changes are positive correlated with the headline changes of article i, which corroborates with our correspondence with the NYT senior editor.[30] Also, the number of headline changes in the same section (columns 1 and 3) is more correlated with the article's likelihood of headline change. Therefore, we use the number of headline changes in the same section as our preferred IV in estimating Equation (4).

[30]We use the lag and lead number of other headline changes as a placebo test. Table A6 in the Appendix shows that not only that the lag and lead variables are negatively correlated with the contemporaneous headline change, the economic and statistical significance of the correlations are much smaller as well.

Table 11: Correlation of Headline Change and # Other Contemporaneous Headline Changes

	(1) Immediate (Same Section)	(2) Immediate (All Sections)	(3) A/B Test (Same Section)	(4) A/B Test (All Sections)
Hours on Homepage	-0.000502*** (0.0000178)	-0.000504*** (0.0000178)	-0.000927*** (0.0000198)	-0.000927*** (0.0000198)
# Headline Changes (Other Articles)	0.0163*** (0.000511)	0.00452*** (0.000173)	0.0102*** (0.000504)	0.00426*** (0.000169)
Observations	498993	498993	498993	498993
Hour FE	Yes	Yes	Yes	Yes
Article FE	Yes	Yes	Yes	Yes
R^2	0.01	0.01	0.01	0.01
F-Stat	110.78	97.15	123.09	132.06

Standard errors in parentheses
* $p < .1$, ** $p < .05$, *** $p < .01$

Table 12 reports the impact of headline changes on various outcome variables using the number of other headline changes in the same section at $t-1$ (IV_{it-1}) as our instrumental variable.

Overall, in the hour after an immediate headline change, the article would become more popular. In particlar, Table 12a shows that there are 43 more tweets sharing the article (0.67 standard deviation increase) after the headline change. The number of positive and negative tweets also increase proportionally compared to before the headline change (approximately 7% and 9% of total tweets). Compared to our results from the PSM estimator in Table 8a , the pro-Democrat tweets increase by a greater proportion (85%) than the average (62%). In other words, instead of headine changes having no additional impact on political slants as suggested by our PSM estimates, our IV approach yields a result that indicates the headline change draws more slanted tweets than before.

Table 12: Impact of Headline Changes (IV)

(a) Immediate Change

	(1) # Tweets (All)	(2) # Tweets (Negative)	(3) # Tweets (Positive)	(4) # Tweets (Dem.)
IV estimate	43.44*** (9.575)	4.849*** (1.536)	5.888*** (1.228)	34.05*** (6.694)
Hours on Homepage	-0.251*** (0.00806)	-0.0319*** (0.00130)	-0.0273*** (0.00104)	-0.163*** (0.00564)
Observations	448884	448884	448884	448884
Hour FE	Yes	Yes	Yes	Yes
Article FE	Yes	Yes	Yes	Yes
F-Stat	22192.13	14717.62	16261.06	19787.72

Standard errors in parentheses
* $p < .1$, ** $p < .05$, *** $p < .01$

(b) A/B Tests

	(1) # Tweets (All)	(2) # Tweets (Negative)	(3) # Tweets (Positive)	(4) # Tweets (Dem.)
IV estimate	32.11** (13.56)	7.232*** (2.198)	3.239* (1.732)	21.03** (9.450)
Hours on Homepage	-0.243*** (0.0137)	-0.0278*** (0.00224)	-0.0273*** (0.00176)	-0.160*** (0.00958)
Observations	448884	448884	448884	448884
Hour FE	Yes	Yes	Yes	Yes
Article FE	Yes	Yes	Yes	Yes
F-Stat	22260.45	14471.83	16433.07	19974.43

Standard errors in parentheses
* $p < .1$, ** $p < .05$, *** $p < .01$

Table 12b reports the impact of A/B tests on various outcome variables. Similar to the impact of an immediate headline change, an A/B test would lead to an increase in popularity of the article. The number of tweets share would increase by 32 (a 0.55 standard deviation increase) in the hour after an A/B test. Similar to our IV results on immediate headline change, an A/B test also seems to increase the level of polarisation. While the the types of readers that an article attracts and the the proportion of positive tweets do not change after an A/B test, the proportion of negative tweets and slanted tweets increase significantly after an A/B test.

6 Wall Street Journal (Preliminary)

In this section we compare headline changes on the New York Times with the Wall Street Journal (WSJ). The WSJ is the logical comparison to the NYT, as it is roughly comparable in size (though smaller in readership) and is the mirror image in terms of political slant (the NYT being on the left and the WSJ on the right). The analysis covers October 2022 to January 2023 and so is preliminary (we are continuing to collect data and will present more complete results in the next draft).

Aside from seeing if the patterns from the previous section carry over to other media sources, analyzing a second source helps to distinguish between leading models of newspaper behavior. For example, newspapers have a range of political slants. This could reflect the supply side in which newspapers have preference over policies (see review in Gentzkow, Shapiro, and Stone, 2015). Alternatively, the slant could come from the demand side: consumers have preferences over political bias and newspapers set their slant to attract readers (that is to profit maximize, not because of any bias of their own). Mullainathan and Shleifer (2005) show that competition need not eliminate this bias. Since both theories predict divergent newspaper biases in equilibrium, observed biases cannot be used to distinguish between them. Instead we test the demand-side model. We evaluate whether readers of a certain newspaper prefer a certain change in bias. Do New York Times readers prefer leftward shifts of headlines while Wall Street Journal readers prefer rightward shifts?[31]

We scraped the WSJ homepage, `https://www.wsj.com/`, in a similar manner as with

[31] Gentzkow and Shapiro (2010) evaluate the source of slant in a demand-side model though they also briefly test a supply side model. However, their sample excludes national newspapers including the New York Times and Wall Street Journal since their identification follows from characteristics of local markets.

the NYT. In particular the Python scraper uses a randomly generated user-agent and does not accept cookies. Among the differences we needed to run the scraper at a higher frequency (every fifteen seconds as opposed to once a minute) since A/B testing periods were shorter. The WSJ also lacks an API and so we extracted article metadata from the WSJ archive, `https://www.wsj.com/news/archive/years`.

We compared the WSJ and NYT results for the four month period where both were scraped, October 2022 through January 2023. All headlines and article abstracts were scored for sentimentality and political slant using the same VADER algorithm and machine learning approach as before. We also collected all tweets of the articles from the two sources, which we then scored for sentimentality and political slant. There are two items omitted from these data: the Blindspotter measure of twitter user partisanship and the WSJ sample omits opinion articles (both of these will be included in the next draft). We aggregated the data to an hourly observation period, checked for headline changes with an A/B test defined as two or more changes within the hour (a headline change with one or no changes is defined as an immediate change).

Table 13 presents summary statistics for articles of both NYT and WSJ. We further present the statistics of articles in different types similar to Table 1. The bottom row shows both newspapers change headlines for about a quarter of the articles, but the WSJ relies more heavily on A/B tests (17% for the WSJ and 6% for the NYT). The top rows shows the articles tend to be relatively similar. WSJ article headlines and text tend to have a slightly more positive sentiment. And as expected the NYT articles have a more pro-Democrat slant, though the difference is not statistically significant (this is partly due to the absence of opinion articles in the WSJ sample). Articles tend to spend a comparable time on the papers' homepages.

Table 13: Summary NYT and WSJ Articles Oct 2022-Jan 2023

(a) NYT

	A/B Test	Immediate	No Switch	Total
Headline	-0.0904	-0.0898	-0.0343	-0.0470
Sentiment	(0.346)	(0.346)	(0.358)	(0.356)
Abstract	-0.0650	-0.0508	0.0170	0.000642
Sentiment	(0.449)	(0.455)	(0.451)	(0.453)
Headline	0.604	0.595	0.609	0.607
Slant	(0.264)	(0.267)	(0.270)	(0.269)
Abstract	0.572	0.584	0.599	0.595
Slant	(0.222)	(0.218)	(0.212)	(0.214)
Hours on	20.97	17.46	13.35	14.51
Homepage	(13.26)	(12.07)	(10.86)	(11.46)
N	602	1551	7308	9461

Mean coefficients; SD in parentheses

(b) WSJ

	A/B Test	Immediate	No Switch	Total
Headline	-0.0521	-0.0334	-0.0223	-0.0290
Sentiment	(0.314)	(0.322)	(0.331)	(0.327)
Abstract	-0.0223	-0.00389	0.0326	0.0178
Sentiment	(0.433)	(0.437)	(0.423)	(0.427)
Headline	0.565	0.553	0.542	0.548
Slant	(0.275)	(0.283)	(0.281)	(0.280)
Abstract	0.538	0.539	0.536	0.537
Slant	(0.216)	(0.214)	(0.207)	(0.210)
Hours on	19.21	17.08	15.75	16.54
Homepage	(10.56)	(10.39)	(13.69)	(12.82)
N	1189	1048	4745	6982

Mean coefficients; SD in parentheses

Table 14 summarizes the Twitter data (all but the top row are hourly values). Similar to the article-level summary statistics, the two newspapers are relatively comparable. NYT articles are tweeted on average three times as often as WSJ ones. The tweets sentimentality (rows three and four) is similar, with NYT articles attracting slightly more negative tweets than WSJ articles. Not surprisingly, the share of tweets that slant to Democrats is larger for the NYT (approximately 60%, which is the last row divided by the second row) than WSJ (43%). Similar to our main NYT sample, articles (of both newspapers) with headline changes are tweeted more frequently than the average article. The sentimentality and partisan slants in the last three rows are comparable to the overall values for both newspapers.

Table 14: Summary NYT and WSJ Tweets Oct 2022-Jan 2023

(a) NYT

	A/B Test	Immediate	No Switch	Total
# Tweets	431.6	397.3	188.9	238.5
(Total)	(1622.8)	(1203.6)	(1161.6)	(1206.4)
# Tweets	24.89	23.07	12.47	15.00
	(99.44)	(61.36)	(51.30)	(57.44)
# Tweets	2.684	2.389	1.191	1.483
(Neg)	(11.94)	(9.326)	(6.775)	(7.685)
# Tweets	2.191	1.912	1.054	1.267
(Pos)	(10.05)	(6.209)	(5.729)	(6.183)
# Tweets	15.83	13.72	7.434	8.999
(Democrat)	(68.98)	(38.96)	(31.13)	(36.17)

Mean coefficients; SD in parentheses

(b) WSJ

	A/B Test	Immediate	No Switch	Total
# Tweets	164.3	114.3	46.09	76.46
(Total)	(473.1)	(340.4)	(163.4)	(275.3)
# Tweets	8.523	6.390	3.570	4.837
	(23.47)	(17.42)	(8.446)	(13.84)
# Tweets	0.674	0.621	0.286	0.402
(Neg)	(1.948)	(2.586)	(1.016)	(1.543)
# Tweets	0.732	0.492	0.303	0.404
(Pos)	(2.573)	(1.293)	(0.950)	(1.420)
# Tweets	3.674	2.708	1.553	2.088
(Democrat)	(9.322)	(7.974)	(3.893)	(5.942)

Mean coefficients; SD in parentheses

6.1 Propensity Score Matching

We estimate the propensity score matching estimator in Section 5.2.1 on our NYT and WSJ sample from Oct 2022 to Jan 2023. The results on the NYT sample is qualitatively consistent with the results in Table 8, with the impact of immediate headline change (Table 15a) essentially the same. The impact of A/B tests in the NYT sample, however, is almost three times as big as those in Table 15c.[32] In both cases, the impact tend to lead to similar level of sentiment polarization and more pro-Democratic tweets in the hour after the headline changes, which are consistent with the results in Table 8.

Table 15: Impact of Headline Changes, NYT vs WSJ (PSM)

(a) NYT: Immediate Change

	(1) # Tweets (All)	(2) # Tweets (Negative)	(3) # Tweets (Positive)	(4) # Tweets (Dem.)
ATE	5.481***	0.483**	0.461**	3.505***
	(1.822)	(0.212)	(0.201)	(1.119)
Observations	117623	117623	117623	117623
Hour FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes

Standard errors in parentheses
* $p < .1$, ** $p < .05$, *** $p < .01$

(b) WSJ: Immediate Change

	(1) # Tweets (All)	(2) # Tweets (Negative)	(3) # Tweets (Positive)	(4) # Tweets (Dem.)
ATE	3.413***	0.477**	0.306***	1.758***
	(0.826)	(0.230)	(0.0657)	(0.579)
Observations	94869	94869	94869	94869
Hour FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes

Standard errors in parentheses
* $p < .1$, ** $p < .05$, *** $p < .01$

(c) NYT: A/B Tests

	(1) # Tweets (All)	(2) # Tweets (Negative)	(3) # Tweets (Positive)	(4) # Tweets (Dem.)
ATE	14.79**	1.567*	1.534**	10.25**
	(7.025)	(0.855)	(0.708)	(4.539)
Observations	117077	117077	117077	117077
Hour FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes

Standard errors in parentheses
* $p < .1$, ** $p < .05$, *** $p < .01$

(d) WSJ: A/B Tests

	(1) # Tweets (All)	(2) # Tweets (Negative)	(3) # Tweets (Positive)	(4) # Tweets (Dem.)
ATE	7.002***	0.775**	0.719**	2.483***
	(2.382)	(0.360)	(0.310)	(0.660)
Observations	95386	95386	95386	95386
Hour FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes

Standard errors in parentheses
* $p < .1$, ** $p < .05$, *** $p < .01$

Headline changes also have a significant impact on the outcome variables for the WSJ. In absolute terms, the impacts are smaller for the WSJ compared to the NYT (roughly 50%). However, given that the number of tweet shares in WSJ is only one third of that in NYT, the relative impact of headline changes is bigger in WSJ.

Interesting, WSJ's tweets became more polarized after both types of headline changes. In particular, there was a large upswing in negative tweets (more than 10%, Column 2 in Tables 15b and 15d). The impact on political slants is mixed: half of the additional tweets after an immediate change were pro-Democrat tweets (Column 4 in 15b) which is more than the average for WSJ articles (approximately 40%), while only 34% of the additional

[32] Table A7 in the Appendix reports the results from the PSM ATET estimator. The results are smaller in magnitude and statisticial significance, but qualitatively similar to the main results.

tweets after an A/B test were pro-Democrat tweets. It seems to suggest while the WSJ editors attempt to attract more readers from outside their core political readership, the A/B tests were doing the opposite.

6.2 Instrumental-Variable Approach

The key results for testing the demand-side model of media bias are Table 16, which are the second stage estimates comparable to those from the last section (the omitted first stage estimates show the positive relationship between other contemporaneous headline changes and own headline changes). While the headline change typically results in more tweets in the next hour (first column), the kind of reader response is somewhat surprising. The last column shows the WSJ is actually getting a large upswing in Democrat-oriented tweets, accounting for a third to a half of the additional tweets (the WSJ headlines also have a more Democrat slant on average after they are changed). In contrast the NYT is getting fewer pro-Democrat tweets. That is neither newspaper is using headline changes to create support among their core political readership. The next draft will have additional details on this point through the Blindspotter data which measure the partisanship of each Twitter account as well as whether they typically consume news from each of these papers.

Table 16: Impact of Headline Changes, NYT vs WSJ (IV)

(a) NYT: Immediate Change

	(1) # Tweets (All)	(2) # Tweets (Negative)	(3) # Tweets (Positive)	(4) # Tweets (Dem.)
IV estimate	6.852	-2.360	3.576	-3.130
	(23.26)	(3.290)	(3.356)	(14.03)
Hours on Homepage	-0.189***	-0.0255***	-0.0192***	-0.134***
	(0.0237)	(0.00336)	(0.00346)	(0.0143)
Observations	129032	129032	129032	129032
Hour FE	Yes	Yes	Yes	Yes
Article FE	Yes	Yes	Yes	Yes
F-Stat	7927.38	5548.39	9778.03	6628.87

Standard errors in parentheses
* $p < .1$, ** $p < .05$, *** $p < .01$

(b) WSJ: Immediate Change

	(1) # Tweets (All)	(2) # Tweets (Negative)	(3) # Tweets (Positive)	(4) # Tweets (Dem.)
IV estimate	-13.63	0.953	-0.978	-3.394
	(11.36)	(1.864)	(1.262)	(6.051)
Hours on Homepage	-0.0783***	-0.00731***	-0.00762***	-0.0388***
	(0.00987)	(0.00167)	(0.00113)	(0.00527)
Observations	104559	104559	104559	104559
Hour FE	Yes	Yes	Yes	Yes
Article FE	Yes	Yes	Yes	Yes
F-Stat	4034.57	1473.52	3995.39	2021.37

Standard errors in parentheses
* $p < .1$, ** $p < .05$, *** $p < .01$

(c) NYT: A/B Tests

	(1) # Tweets (All)	(2) # Tweets (Negative)	(3) # Tweets (Positive)	(4) # Tweets (Dem.)
IV estimate	-4.089	-0.262	-0.717	1.918
	(11.53)	(1.629)	(1.655)	(6.954)
Hours on Homepage	-0.195***	-0.0237***	-0.0223***	-0.131***
	(0.0144)	(0.00204)	(0.00207)	(0.00869)
Observations	129032	129032	129032	129032
Hour FE	Yes	Yes	Yes	Yes
Article FE	Yes	Yes	Yes	Yes
F-Stat	7932.30	5560.92	9847.40	6629.99

Standard errors in parentheses
* $p < .1$, ** $p < .05$, *** $p < .01$

(d) WSJ: A/B Tests

	(1) # Tweets (All)	(2) # Tweets (Negative)	(3) # Tweets (Positive)	(4) # Tweets (Dem.)
IV estimate	86.45***	4.367	4.241	31.02**
	(33.08)	(4.016)	(2.842)	(14.92)
Hours on Homepage	-0.00638	-0.00475	-0.00366*	-0.0139
	(0.0242)	(0.00315)	(0.00218)	(0.0110)
Observations	104559	104559	104559	104559
Hour FE	Yes	Yes	Yes	Yes
Article FE	Yes	Yes	Yes	Yes
F-Stat	1987.23	1346.90	3367.65	1381.77

Standard errors in parentheses
* $p < .1$, ** $p < .05$, *** $p < .01$

7 Conclusion

Using a PSM and an IV approach, we find that headline changes increase an article's popularity. There are some evidence that headline changes increase the article's tweets' negative sentiment and liberal political slant. These results are suggestive that economic motives are one of the main driving forces behind the New York Times headline change strategy. This has significant implications for media competition and regulation, especially in the age of digitization which is often argued amplifies falsehoods and creates echo chambers.

A significant challenge for our research is fully understanding the economic benefits of such a strategy. While we can observe more engagement on Twitter and higher popularity on the paper's website when headline changes or testing occurs, it is not clear how this translates into revenue. As discussed in the introduction, the New York Times is collecting a growing shares of its revenue from subscription rather than advertising. Does the engagement we document here translate into subscriptions? That is something we are unable to analyze here, and of course it is a more general issue applicable to other growing subscription platforms such as Netflix or Spotify.

Finally, we note an ethical issue. The New York Times is among the small group of newspapers of record. Its stories are supposed to capture the most important news, and how they interpreted at the moment they occur. But newspaper articles are no longer set in amber when headlines change. Perhaps more worrisomely, the newspaper does not alert the reader to the headline changes nor if there were experiments. This creates the possibility of a memory hole (Orwell, 1949) as headlines at least are written to reflect current beliefs or even for more nefarious purposes. Presumably listing headline changes on the bottom of each article would be a relativity costless step that would mitigate this possibility.

References

Allcott, Hunt and Matthew Gentzkow. 2017. “Social Media and Fake News in the 2016 Election.” *Journal of Economic Perspectives* 31 (2):211–236.

Allen, Jennifer, Baird Howland, Markus Mobius, David Rothschild, and Duncan J. Watts. 2020. “Evaluating the fake news problem at the scale of the information ecosystem.” *Science Advances* 6 (14):eaay3539.

Anderson, Monica, Lee Rainie, and Haley Nolan. 2021. “Social Media Use in 2021.” Tech. rep., Pew Research Center.

Angelucci, Charles and Julia Cagé. 2019. “Newspapers in Times of Low Advertising Revenues.” *American Economic Journal: Microeconomics* 11 (3):319–364.

Angelucci, Charles, Julia Cagé, and Michael Sinkinson. 2020. “Media Competition and News Diets.” NBER Working Paper 26782.

Arak, Josh and Kentaro Kaji. 2017. “ABRA: An enterprise framework for experimentation at The Times.” *New York Times* URL `https://open.nytimes.com/abra-an-enterprise-framework-for-experimentation-at-the-times-57f8931449cd`.

Athey, Susan and Michael Luca. 2019. “Economists (and Economics) in Tech Companiess.” *Journal of Economic Perspectives* 33 (1):209–230.

Baum, Matthew A. and Tim Groeling. 2008. “New Media and the Polarization of American Political Discourse.” *Political Communication* 25 (4):345–365.

Bird, Steven, Ewan Klein, and Edward Loper. 2009. *Natural Language Processing with Python*. O’Reilly Media Inc.

Boxell, Levi and Jacob Conway. 2022. “Journalist Ideology and the Production of News: Evidence from Movers.”

Boxell, Levi, Matthew Gentzkow, and Jesse M Shapiro. 2018. “A note on internet use and the 2016 U.S. presidential election outcome.” *PLOS ONE* 13 (7):1–7.

Bulik, Mark. 2016. “Which Headlines Attract Most Readers?” *New York Times* URL `https://www.nytimes.com/2016/06/13/insider/which-headlines-attract-most-readers.html`.

Cagé, Julia, Moritz Hengel, Nicolas Hervé, and Camille Urvoy. 2022. "Hosting Media Bias: Evidence from the Universe of French Broadcasts, 2002-2020." URL https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4036211.

Cagé, Julia, Nicolas Hervé, and Béatrice Mazoyer. 2022. "Social Media and Newsroom Production Decisions." URL https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3663899.

Campante, Filipe, Ruben Durante, and Francesco Sobbrio. 2018. "Politics 2.0: The Multifaceted Effect of Broadband Internet on Political Participation." *Journal of the European Economic Association* 16 (4):1094–1136.

Claussen, Jorg, Christian Peukert, and Ananya Sen. 2020. "The Editor and the Algorithm: Returns to Data and Externalities in Online News." URL https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3518959.

de Corniére, Alexandre and Miklos Sarvary. 2022. "Social Media and News: Content Bundling and News Quality." *Management Science* :forthcoming.

Ecker, Ullrich K H, Stephan Lewandowsky, Ee Pin Chang, and Rekha Pillai. 2014. "The effects of subtle misinformation in news headlines." *Journal of Experimental Psychology: Applied* 20 (4):323–335.

Falck, Oliver, Robert Gold, and Stephan Heblich. 2014. "E-lections: Voting Behavior and the Internet." *American Economic Review* 104 (7):2238–2265.

Gans, Joshua S., Avi Goldfarb, and Mara Lederman. 2021. "Exit, Tweets, and Loyalty." *American Economic Journal: Microeconomics* 13 (2):68–112.

Gentzkow, Matthew. 2006. "Television and Voter Turnout." *The Quarterly Journal of Economics* 121 (3):931–972.

Gentzkow, Matthew, Edward L. Glaeser, and Claudia Goldin. 2006. *The Rise of the Fourth Estate. How Newspapers Became Informative and Why It Mattered*. University of Chicago Press, 187–230. URL http://www.nber.org/chapters/c9984.

Gentzkow, Matthew and Jesse M. Shapiro. 2008. "Competition and Truth in the Market for News." *Journal of Economic Perspectives* 22 (2):133–154.

———. 2010. “What Drives Media Slant? Evidence from U.S. Daily Newspaper.” *Econometrica* 78 (1):35–71.

Gentzkow, Matthew, Jesse M. Shapiro, and Michael Sinkinson. 2011. “The Effect of Newspaper Entry and Exit on Electoral Politics.” *American Economic Review* 101 (7):1980–3018.

Gentzkow, Matthew, Jesse M. Shapiro, and Daniel F. Stone. 2015. “Media Bias in the Marketplace: Theory.” In *Handbook of Media Economics, Volume 1*, edited by Simon P. Anderson, Joel Waldfogel, and David Strömberg. Amsterdam: Elsevier, 623–645.

George, Lisa M. and Joel Waldfogel. 2006. “The New York Times and the Market for Local Newspapers.” *American Economic Review* 96 (1):435–447.

Granato, Len. 2002. *Newspaper Feature Writing*. University of New South Wales Press in association with Deakin University Press.

Groseclose, Tim and Jeffrey Milyo. 2005. “A Measure of Media Bias*.” *The Quarterly Journal of Economics* 120 (4):1191–1237.

Hagar, Nick and Nicholas Diakopoulos. 2019. “Optimizing Content with A/B Headline Testing: Changing Newsroom Practices.” *Media and Communication* 7 (1):117–127.

Hatte, Sophie, Etienne Madinier, and Ekaterina Zhuravskaya. 2022. “Reading Twitter in the Newsroom: Web 2.0 and Traditional-Media Reporting of Conflicts.” URL `https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3845739`.

Hirano, Shigeo and James M. Snyder. 2020. “Measuring the Partisan Behavior of U.S. Newspapers, 1880 to 1980.”

Huang, Yihong, Juanjuan Meng, and Xi Weng. 2022. “Strategic Thinking and Media Bias: Evidence from Chinese Microblog Users.” URL `https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4184003`.

Hutto, Clayton and Eric Gilbert. 2014. “Vader: A parsimonious rule-based model for sentiment analysis of social media text.” *Proceedings of the international AAAI conference on web and social media* 8 (1):216–225.

Jeon, Doh-Shin and Nikrooz Nasr. 2016. "News Aggregators and Competition among Newspapers on the Internet." *American Economic Journal: Microeconomics* 8 (4):91–114.

Jiang, Shan, John Martin, and Christo Wilson. 2019. "Who's the Guinea Pig? Investigating Online A/B/n Tests in-the-Wild." In *Proceedings of the 2019 ACM Conference on Fairness, Accountability, and Transparency (FAccT 2019).*

Konnikova, Maria. 2014. "How Headlines Change the Way We Think." *New Yorker* .

Locke, Julian and Sameer More. 2021. "How We Rearchitected Mobile A/B Testing at The New York Times." *New York Times* .

Martin, Gregory J. and Joshua McCrain. 2019. "Local News and National Politics." *American Political Science Review* 113 (2):372–384.

Martin, Gregory J. and Ali Yurukoglu. 2017. "Bias in Cable News: Persuasion and Polarization." *American Economic Review* 107 (9):2565–99.

Muise, Daniel, Homa Hosseinmardi, Baird Howland, Markus Mobius, David Rothschild, and Duncan J. Watts. 2022. "Quantifying partisan news diets in Web and TV audiences." *Science Advances* 8 (28):eabn0083.

Mullainathan, Sendhil and Andrei Shleifer. 2005. "The Market for News." *American Economic Review* 95 (4):1031–1053.

New York Post. 2007. *Headless Body in Topless Bar: The Best Headlines from America's Favorite Newspaper.* HarperCollins.

Orwell, George. 1949. *Nineteen Eighty-Four.* Secker & Warburg.

Qin, Bei, David Strömberg, and Yanhui Wu. 2018. "Media Bias in China." *American Economic Review* 108 (9):2442–2476.

Rajkumar, Karthik, Guillaume Saint-Jacques, Iavor Bojinov, Erik Brynjolfsson, and Sinan Aral. 2022. "A causal test of the strength of weak ties." *Science* 377 (6612):1304–1310.

Ray, Bonnie. 2022. "Personal correspondence." `https://chartbeat.com/`.

Reisman, Dillon. 2016a. "All the News That's Fit to Change: Insights into a corpus of 2.5 million news headlines." URL https://freedom-to-tinker.com/2016/09/14/all-the-news-thats-fit-to-change-insights-into-a-corpus-of-2-5-million-news-head

———. 2016b. "A Peek at A/B Testing in the Wild." URL https://freedom-to-tinker.com/2016/05/26/a-peek-at-ab-testing-in-the-wild/.

Shapiro, Jesse M. 2016. "Special interests and the media: Theory and an application to climate change." *Journal of Public Economics* 144:91–108.

Snyder, James M and David Strömberg. 2010. "Press Coverage and Political Accountability." *Journal of Political Economy* 118 (2).

Strömberg, David. 2004. "Radio's Impact on Public Spending." *The Quarterly Journal of Economics* 119 (1).

Symonds, Alexandria. 2017. "When a Headline Makes Headlines of Its Own." *New York Times* URL https://www.nytimes.com/2017/03/23/insider/headline-trump-time-interview.html.

Wang, Tianyi. 2022. "Waves of Empowerment: Black Radio and the Civil Rights Movement."

Appendix A Challenges in Defining A/B Tests

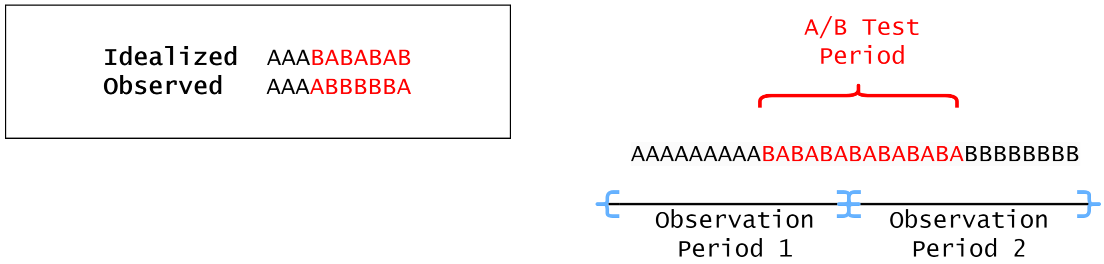
In this section we discuss the two challenges of defining A/B tests in our sample.

The first challenge to detecting these trials is that we observe a random sample of headlines during an A/B test.This means we may not detect the test until part way through the experiment period as we might observe few headline changes. In some cases we may fail to detect an A/B test. Figure A1a illustrates the issue.

Figure A1: Measurement Issues with A/B tests (red text indicates an A/B test)

(a) Observation Issue

(b) A/B Test Straddle Observation Periods



How likely are we to miss a test? Suppose that during an A/B test the NYT randomly shows the initial headline type A with probability p to a reader and headline type B with probability $1 - p$. Further suppose our scrapes receive independent draws from these distributions (since each scrape uses a random user agent) and the testing period lasts for N scrapes. In our NYT application, $p = 0.5$ and $N = 30$.

To derive the probability of misclassifying an A/B test, note that headline changes follow a binomial distribution,[33]

$$\Pr(k \text{ headline changes during } N \mid \text{A/B test}) = \binom{N}{k} p^k (1-p)^{N-k} \tag{6}$$

The two possible misclassifications are that we never observe a headline change or only observe an immediate change. We can write formulae for these,

$$\begin{gathered} \Pr(\text{misclassified as no headline change}) \equiv \\ \Pr(0 \text{ headline changes observed} \mid \text{A/B test}) = (1-p)^N \end{gathered} \tag{7}$$

[33]The formula accounts for the prior period using headline A, so a B headline in the first scrape would be considered a headline change. We ignore a headline change at the start of a period following an A/B test (e.g. the test ends with an observation of type A and the next period starts with the winning headline B.

and

$$
\begin{aligned}
&\Pr(\text{misclassified as immediate headline change}) \equiv \\
&\quad \Pr(\text{1 headline changes observed} \mid \text{A/B test}) = Np(1-p)^{N-1}
\end{aligned} \tag{8}
$$

Both of these terms are quite small in our application: Pr(misclassified as no headline change) = $2^{-30} \approx 10^{-9}$ and Pr(misclassified as immediate headline change) = $30 \times 2^{-30} \approx 10^{-8}$. In short we are unlikely to misclassify an A/B test.

This approach abstracts from an important issue: we will analyze the data in short time chunks, and these observation periods may not perfectly align with the newspaper's testing period (for example, the test starts on the tenth scrape).[34] The formulae still hold unless the test period straddles adjacent observation periods as illustrated in Figure A1b, which lead to two complications. First, it can lead to a single A/B test being detected in two adjacent observation periods.[35] This is minor issue since most of the analysis focuses on whether there is any A/B test at all, and even when we focus on which periods have a test we are correctly classifying our observation periods. Second, we need to modify the formulae for when we do not detect an A/B test. For simplicity we combine the two types of misclassification,

$$
\begin{aligned}
\Pr(\text{misclassified when A/B test straddles}) &\equiv \\
\Pr(\text{not detect A/B in OP=1}) \times \Pr(\text{not detect A/B in OP=2}) &= \\
(1-p)^{N_1-1}[(1-p)+N_1p] \times (1-p)^{N_2-1}[(1-p)+N_2p] &= \\
(1-p)^{T-2}[(1-p)+N_1p][(1-p)+(T-N_1)p] &=
\end{aligned} \tag{9}
$$

where $OP = i$ is the ith observation period where the test occurs, N_i is the number of scrapes in $OP = i$ where there is an A/B test running, and the last line uses $N_1 + N_2 = T$. To be conservative we will look for the worst case, where this misclassification probability is as large as possible. Some calculus shows this function has a maximum at $N_1 = T/2$ (the A/B test is evenly split between the observation periods) which leads to,

[34] The length of the observation period may also differ from the A/B test. We presume the observation period is at least as long as the A/B test period, which means the test can be in no more than two observation periods.

[35] Pr(Two adjacent A/B tests observed when A/B test straddles) = $1-(1-p)^{T-N_1-1}[1-p+(T-N_1)p]-(1-p)^{N_1-1}[1-p+N_1p]+(1-p)^{T-2}[1-p+N_1p][1-p+(T-N_1)p]$ using the notation in the next paragraph. This probability is near zero when $N_1 \to T$ or $N_1 \to 0$ (almost all of the A/B test is in one of the observation periods) and near one when $N_1 \to T/2$ (the test is evenly split between the observation periods).

$$\max_{N_1} \Pr(\text{misclassified when A/B test straddles}) = (1-p)^{T-2}(1-p+Tp/2)^2 \quad (10)$$

Using values for our application, $\max \Pr(\text{misclassified when A/B test straddles}) \approx 10^{-7}$. Even in this worst case scenario, this is still rather small.

Appendix B Cannibalization

An important concern is that headline changes do not expand consumption as we presume in the text but instead just reallocate consumption from other articles. If there is such *cannibalization* than headline changes would have little economic benefit to the newspaper. In this section we show that the greater attention articles receive after a headline change does not come at the cost of reduced popularity of other articles.[36]

If there is cannibalization, than it should presumably be strongest for related articles: if people increase reading a business article, than other business articles covering similar topics and published at a similar time should see the greatest reduction in readership (as with the main text, we will use tweet activity as a proxy for readership).

We use a standard bag-of-words approach to determine article similarity. The NYT API labels each articles with a set of tags ("TimesTags"). On average there are 7.4 tags per article and 23.8k unique tags. We use *Jaccard similarity* to measure the similarity between the set of tags for any two articles (A, B),

$$J_{AB} \equiv \frac{|A \cap B|}{|A \cup B|} \in [0, 1] \tag{11}$$

The Jaccard index measure the overlap in tags for the articles and higher values indicate greater overlap (zero means no overlap, one perfect overlap).

We calculate J for every pair of headlines (A, B), and than limit the sample to pairs where A has headline change (A/B or immediate) and B is active in the hour of change. For reference, for articles with headline changes 16% article-hours have $J > 0$ and $J_{ave}|J > 0 = 0.10$.

The specification we consider is,

$$Popularity_{it} = \beta J_{it}^{Headline\,\Delta} + controls + \nu it \tag{12}$$

This estimates whether article i's popularity in hour t is diminished if some other article has a headline change, and scales this effect by how similar the articles are. If there is significant cannibalization that $\beta < 0$. Since there are multiple headline changes in most hours, we discuss a below a variety of ways of defining the similarity index. The dependent variable is number of tweets in the hour after the headline change (the results

[36]The analysis is restricted to data collected over Febraaury-June 2021.

are qualitatively similar when we use changes in the hour of the change).

Table A1: Testing for Cannibalization after Headline Change

	(1) # Tweets ($I(change\ in\ t)_{it}$)	(2) # Tweets ($\sum I(change\ in\ t)_{it}$)	(3) # Tweets ($\overline{J}_{it}$)	(4) # Tweets ($\sum J_{it}$)
Similarity	0.2705	0.43640	0.8163	2.0627
	(0.56794)	(0.40619)	(4.12196)	(3.09990)
Observations	163128	163128	163128	163128
Hours on Homepage	Yes	Yes	Yes	Yes
Hour FE	Yes	Yes	Yes	Yes
Article FE	Yes	Yes	Yes	Yes
F-Stat	82.19	82.23	82.19	82.20

Similarity measure listed in parenthesis at top of column
Standard errors in parentheses
* $p < .1$, ** $p < .05$, *** $p < .01$

Table A1 reports the estimates. The columns headers list the similarity measure used in that specification: (1) uses an indicator if there is any other related article ($J_{i.} > 0$) with a headline change in that hour, (2) is the number of other related articles with a headline change in that hour, (3) is the average Jaccard index among related articles with a headline change in that hour, and (4) is the sum of the Jaccard indices among related articles with a headline change in that hour. There is no evidence of cannibalization: the $\hat{\beta}$ estimates are positive (though not statistically or economically significant).

Appendix C PSM: Transition Matrix of Tweet Counts

In addition to comparing the number of tweet shares between the treatment and control groups, we also compare the upward/downward mobility of tweet shares between the two groups. In particular, we discretize the number of tweet shares in six groups and construct the transition matrix of tweet shares from the hour of the headline change to the hour after the headline change, for the two groups separately.The six tweet shares group are defined as follows: Group 0 has zero tweet; Group 1 has between 1 to 2 tweets (which is the 30th percentile of the tweet shares distribution); Group 2 has between 3 to 8 tweets (60th percentile); Group 3 has between 9 to 22 tweets (80th percentile); Group 4 has between 23 to 46 tweets (90th percentile); and Group 5 has more than 46 tweets.[37]

Table A2 shows the tweet shares' transition matrix for the control group (left panel) and the treatment group (right panel), respectively. The results suggest that an immediate headline change tend to decrease downward mobility and increase upward mobility of tweet shares. The effects tend to be stronger for articles with between 1 to 22 tweet shares (group 1, 2, and 3) at the hour of the headline change. For instance, if an article had between 1 to 2 tweets at the hour of headline change, the probability that it would move up to a higher tweet share group (2 or above) is approximately 43%, compared to 37% of the articles in the control group. The probability that it would move down to group 0, i.e. zero tweet, in the next hour would be 23%, compared to 24% in the control group.

Table A2: Transition Matrix of Tweets (Immediate Change vs Control)

(a) Control Group

	0	1	2	3	4	5
0	.604	.246	.103	.032	.009	.007
1	.242	.39	.296	.058	.01	.004
2	.065	.23	.461	.191	.038	.015
3	.006	.048	.314	.462	.129	.04
4	.002	.004	.071	.376	.392	.156
5	0	.002	.005	.063	.213	.716

(b) Immediate Headline Change

	0	1	2	3	4	5
0	.623	.218	.096	.044	.009	.011
1	.227	.341	.33	.082	.014	.006
2	.072	.211	.431	.218	.046	.022
3	.008	.05	.319	.44	.132	.051
4	.001	.007	.07	.382	.382	.158
5	0	.001	.006	.065	.219	.709

However, the upward mobility of the lowest group (i.e. those with zero tweet at the hour of headline change) and the downward mobility of the highest group (i.e. those with more than 46 tweets at the hour of headline change) are essentially the same between the

[37]The results are robust to different cutoff points for the tweet shares group.

control and treatment group. Overall, the results in Table A2 suggest that immediate headline change would help most of the articles in the sample.

Table A3 shows a very similar picture about the impact of A/B tests on tweet shares' upward/downward mobility, where the impact is the biggest among groups 2 to 4 (articles with between 1 to 22 tweet shares). For instance, an article undergone an A/B test in group 1 (between 1 to 2 tweets at the hour of an A/B test) would have 40% chance of moving up to group 2 or above, while an article in the control group would have 35% chance of doing so.

Table A3: Transition Matrix of Tweets (A/B Tests vs Control)

(a) Control Group

	0	1	2	3	4	5
0	.624	.247	.096	.019	.006	.008
1	.274	.376	.285	.049	.011	.005
2	.071	.233	.49	.172	.026	.008
3	.009	.045	.309	.476	.123	.038
4	0	.005	.073	.385	.38	.157
5	0	.001	.004	.053	.198	.744

(b) A/B Tests

	0	1	2	3	4	5
0	.623	.225	.104	.026	.01	.012
1	.226	.374	.334	.053	.012	.002
2	.062	.225	.474	.196	.032	.011
3	.007	.044	.293	.485	.131	.041
4	.001	.003	.077	.381	.395	.143
5	0	0	.004	.049	.188	.759

Appendix D Additional Tables

Table A4: Impact of Headline Changes (PSM, ATET)

(a) Immediate Change

	(1) # Tweets (All)	(2) # Tweets (Negative)	(3) # Tweets (Positive)	(4) # Tweets (Dem.)
ATET	5.911***	0.501*	0.520***	3.781***
	(1.480)	(0.275)	(0.138)	(1.018)
Observations	400621	400621	400621	400621
Hour FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes

Standard errors in parentheses
* $p < .1$, ** $p < .05$, *** $p < .01$

(b) A/B Tests

	(1) # Tweets (All)	(2) # Tweets (Negative)	(3) # Tweets (Positive)	(4) # Tweets (Dem.)
ATET	3.202**	0.435**	0.260*	1.750**
	(1.278)	(0.192)	(0.148)	(0.802)
Observations	403583	403583	403583	403583
Hour FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes

Standard errors in parentheses
* $p < .1$, ** $p < .05$, *** $p < .01$

Table A5: Impact of Headline Changes on Tweet Shares by Tone (PSM, ATET)

(a) Immediate Change

	(1) Feature	(2) News	(3) Opinion
ATET	2.113	8.189***	-0.773
	(1.723)	(2.014)	(2.819)
Observations	96766	193703	110152
Hour FE	Yes	Yes	Yes
Month FE	Yes	Yes	Yes

Standard errors in parentheses
* $p < .1$, ** $p < .05$, *** $p < .01$

(b) A/B Tests

	(1) Feature	(2) News	(3) Opinion
ATEE	2.026**	5.954**	-0.0874
	(0.937)	(2.370)	(0.826)
Observations	97369	195301	110913
Hour FE	Yes	Yes	Yes
Month FE	Yes	Yes	Yes

Standard errors in parentheses
* $p < .1$, ** $p < .05$, *** $p < .01$

Table A6: Correlation of Headline Change and # Other (Lag/Lead) Headline Changes

	(1) Non-A/B (Lag)	(2) Non-A/B (Lead)	(3) A/B Test (Lag)	(4) A/B Test (Lead)
Hours on Homepage	-0.000719***	-0.000509***	-0.000505***	-0.000446***
	(0.0000445)	(0.0000461)	(0.0000358)	(0.0000378)
# Headline Changes (Other Articles)	-0.00152	-0.00248**	-0.00197*	-0.00223*
	(0.00103)	(0.00106)	(0.00111)	(0.00117)
Observations	144090	144090	144090	144090
Hour FE	Yes	Yes	Yes	Yes
Article FE	Yes	Yes	Yes	Yes
R^2	0.01	0.01	0.00	0.00
F-Stat	34.09	32.97	9.95	8.94

Standard errors in parentheses

* $p < .1$, ** $p < .05$, *** $p < .01$

Table A7: Impact of Headline Changes, NYT vs WSJ (PSM, ATET)

(a) NYT: Immediate Change

	(1) # Tweets (All)	(2) # Tweets (Negative)	(3) # Tweets (Positive)	(4) # Tweets (Dem.)
ATET	1.759	0.0830	0.587*	0.950
	(3.054)	(0.384)	(0.335)	(1.863)
Observations	117623	117623	117623	117623
Hour FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes

Standard errors in parentheses
* $p < .1$, ** $p < .05$, *** $p < .01$

(b) WSJ: Immediate Change

	(1) # Tweets (All)	(2) # Tweets (Negative)	(3) # Tweets (Positive)	(4) # Tweets (Dem.)
ATET	2.561**	0.514**	0.197**	1.552**
	(1.144)	(0.217)	(0.0902)	(0.625)
Observations	94869	94869	94869	94869
Hour FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes

Standard errors in parentheses
* $p < .1$, ** $p < .05$, *** $p < .01$

(c) NYT: A/B Tests

	(1) # Tweets (All)	(2) # Tweets (Negative)	(3) # Tweets (Positive)	(4) # Tweets (Dem.)
ATET	4.799	0.631	0.394	3.484
	(3.365)	(0.391)	(0.325)	(2.199)
Observations	117077	117077	117077	117077
Hour FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes

Standard errors in parentheses
* $p < .1$, ** $p < .05$, *** $p < .01$

(d) WSJ: A/B Tests

	(1) # Tweets (All)	(2) # Tweets (Negative)	(3) # Tweets (Positive)	(4) # Tweets (Dem.)
ATET	1.536	0.00826	0.128	0.733*
	(0.972)	(0.0924)	(0.106)	(0.436)
Observations	95386	95386	95386	95386
Hour FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes

Standard errors in parentheses
* $p < .1$, ** $p < .05$, *** $p < .01$